The Hidden Cost of the Cloud: Data Centers and Transmission-Induced Electricity Prices*

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Abstract

We estimate the causal effect of concentrated demand growth on transmission-induced electricity prices. Leveraging data center expansion in Virginia (2015–2024) and a propensity score matching difference-in-differences design at the census-tract level, we find that data centers increases transmission-induced wholesale electricity prices by $2.49 per MWh, approximately 70 percent above pre-treatment levels. Price effects stem from spatial misalignment between load and generation, and increase with distance to nearby generation. Effects are larger during peak demand and more intensive load growth, and fall disproportionately on non-White communities, informing efforts to align load siting and generation placement with transmission capacity.

Keywords: Data Centers; Electricity Demand; Transmission Constraint; Distributional Impact

JEL Classification: Q41, Q48, L94, L86, R11

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1 Introduction

Large, geographically concentrated demand shocks are increasingly reshaping electricity markets (EIA, 2026; IEA, 2026). Recent projections suggest that data centers alone could account for roughly 12 percent of total U.S. electricity consumption by 2028 (Shehabi et al., 2024). These expansions are highly localized, often occurring within a small number of communities, yet they operate within integrated regional markets where prices are determined at a broader scale. As a result, the economic effects of local demand growth depend not only on aggregate supply and demand, but also on where demand arises relative to generation and transmission infrastructure.

In a competitive setting without transmission constraints, a demand increase raises the market-clearing price to reflect the marginal cost of generation, and that price applies uniformly across locations within the market. In electricity markets, however, transmission constraints can create spatial wedges between marginal generation costs and local prices. When transmission lines are congested, prices diverge across locations, and consumers near congestion points may face higher prices even if the marginal cost of generation is unchanged. These price differences reflect infrastructure constraints rather than underlying production costs. To the extent that local prices deviate from the uniform marginal generation benchmark because of transmission constraints, concentrated demand growth may generate spatial price wedges.

As concentrated load growth becomes more common, it is not obvious ex ante how it generates spatially differentiated price effects through transmission constraints. Engineering models can simulate system-wide energy costs, but they face fundamental limitations in predicting the spatial distribution of transmission-induced price components. In interconnected transmission systems, a localized demand increase affects flows throughout the network in ways that depend on topology and dispatch conditions.[1] Identifying which communities experience higher prices, and by how much, therefore requires empirical analysis (Fuentes et al., 2025; Hobbie et al., 2022).

The distributional consequences of these spatial price wedges are economically meaningful. When transmission constraints raise prices in some locations but not others, they reshape the geographic incidence of market outcomes within an otherwise integrated system.

[1] In meshed transmission networks, electricity follows the path of least resistance rather than predetermined contractual routes. When localized demand increases and congests direct transmission lines, power automatically shifts through neighboring regions — a phenomenon known as "loop flows." Congestion at one location can therefore overload lines far from the original demand shock, making it difficult to predict which constraints will bind and where congestion prices will emerge. While engineering models can simulate aggregate system costs under assumed operating conditions, they rely on detailed structural inputs and counterfactual dispatch assumptions. Such models are well-suited for system planning, but they do not directly identify the realized causal effect of localized demand growth on spatial price wedges.

Such effects may affect future siting and investment decisions. Measuring their magnitude and spatial extent is therefore central to understanding the economic effects of concentrated demand growth.

This paper estimates the causal effect of concentrated demand shocks on spatially differentiated wholesale electricity prices using the expansion of data centers in Virginia between 2015 and 2024. Wholesale prices in PJM are measured as locational marginal prices (LMPs), which decompose into a uniform system energy component and congestion and marginal loss components. The system energy component reflects the marginal cost of generation across the market, while the congestion and loss components capture deviations induced by transmission constraints. We define the transmission-induced price component as the combined congestion and marginal loss components of LMPs, as they represent deviations from the uniform marginal generation benchmark driven by infrastructure constraints rather than changes in production costs. Estimating the effect of data center expansion on the transmission-induced price component allows us to quantify how concentrated load growth generates spatial price wedges and reshapes the geographic incidence of electricity market outcomes.

Virginia provides a useful empirical setting for three reasons. First, it is the world's largest data center hub, generating substantial variation in the timing and intensity of concentrated load growth (EIA, 2024, 2025). Second, the uniform system energy component in PJM provides a market-wide benchmark for marginal generation costs, so spatial variation in the transmission-induced price component reflects infrastructure constraints rather than differences in underlying production costs. Third, severe interconnection backlogs and limited transmission expansion during this period limit supply-side responses, allowing us to isolate the demand-driven effects of data center expansion (Rand et al., 2024).

We estimate the effect of data center entry using a propensity score–matched difference-in-differences design. We aggregate nodal-level LMP data and detailed data center operational records to the census tract level and leverage staggered entry timing across locations. Propensity score matching balances observable tract characteristics across treated and untreated areas, addressing concerns that data centers are systematically located in demographically or economically distinct communities. We then implement a staggered difference-in-differences design to estimate the causal effect of data center entry on the transmission-induced price component. Our preferred specification follows Sun and Abraham (2021), and we also report estimates from a two-way fixed effects model and Callaway and Sant'Anna (2021). Results are consistent across estimators.

Our main result shows that data center entry increases the transmission-induced price component by $2.49 per megawatt hour (MWh), corresponding to a 70 percent increase

relative to pre-treatment levels. This causal estimate indicates that concentrated load growth substantially increases local spatial price wedges.

We next explore potential mechanisms underlying this effect. The results suggest spatial misalignment between concentrated load growth and nearby generation capacity. The transmission-induced price component increases more when data centers are located farther from generation resources. When generation is co-located within the same census tract, the transmission-induced price component is smaller in areas with fossil fuel generation and larger in areas with renewable generation. In contrast, when generation lies outside the data center's tract, the increase in the transmission-induced price component does not differ by fuel type.

We also find substantial heterogeneity across time, treatment intensity, and demographics. The increase in the transmission-induced price component is particularly pronounced during peak demand hours and in spring, when transmission constraints are most binding. The effect is larger in areas experiencing more intensive data center expansion and in predominantly non-White communities, highlighting the unequal spatial and distributional consequences of concentrated load growth.

These findings have broader implications for infrastructure planning under rapid electrification. The price consequences of concentrated load growth depend critically on spatial coordination between demand and supply capacity. The peak-hour amplification and demographic disparities we document further indicate that rapid data center expansion can intensify existing spatial and distributional pressures. Coordinated investment in proximate generation capacity, together with greater temporal flexibility in large-load consumption, may therefore be important for mitigating transmission-induced price effects associated with concentrated demand growth.

This paper contributes to the literature in three ways. First, it contributes to the literature on infrastructure constraints in electricity markets. Prior work emphasizes transmission constraints (Davis, Hausman and Rose, 2023) and generation investment in response to large loads (Ham, Kay and Hausman, 2025; Hausman, 2025). We add to this literature by examining how large, geographically concentrated demand shocks affect spatial prices when transmission capacity is fixed. Our estimates show that data center entry increases transmission-induced price component by about 70 percent, indicating that demand-side shocks can generate substantial spatial price wedges.

Second, we contribute to the emerging literature on the economic consequences of data center expansion. Existing studies rely on engineering simulations of aggregate consumption (e.g., Masanet et al., 2020, 2011; Norris et al., 2025; Shehabi et al., 2018; Wade et al., 2025; Wilson, Paschalidis and Coskun, 2022; Zhang et al., 2025) or ex-ante economic modeling

(Knittel, Senga and Wang, 2025). While these approaches provide valuable projections of system-wide effects, they are not designed to predict spatially differentiated price outcomes, as transmission-induced price components depend on realized network conditions, dispatch, and topology that are difficult to simulate ex ante. We add to this literature by, to the best of our knowledge, providing the first ex-post causal estimates of how data center entry affects spatially differentiated electricity prices using observed market outcomes.

Third, we contribute to the literature on the distributional implications of energy infrastructure and policies. Prior work documents that racial and socioeconomic groups face disproportionate exposure to environmental burdens from energy production (Currie, Voorheis and Walker, 2023) and that market-based environmental policies can exacerbate or mitigate these disparities (Carley and Konisky, 2020; Davis and Hausman, 2022; Hernandez-Cortes and Meng, 2023). We extend this literature by showing that concentrated data center expansion generates unequal pricing impacts across communities, highlighting the distributional consequences of rapid load growth.

The remainder of the paper proceeds as follows. Section 2 provides institutional background. Section 3 presents the conceptual framework. Section 4 describes the data, and Section 5 outlines the empirical strategy. Section 6 presents the results and robustness analyses. Section 7 concludes with policy implications.

2 Institutional Background and Market Environment

2.1 Data Center Expansion and Policy Environment in Virginia

Virginia is the largest data center hub in the world, with Northern Virginia, often referred to as "Data Center Alley", hosting over one-third of global data center capacity. This concentration reflects extensive fiber infrastructure, proximity to major internet exchange points, competitive electricity prices, and a business-friendly policy environment. As of 2025, data center expansion in the region accounts for more than 90 percent of projected new electricity demand in PJM's Virginia service territory (Reuters, 2025). Nationally, data centers consumed approximately 4.4 percent of total U.S. electricity in 2023, a figure projected to rise substantially over the next decade (Shehabi et al., 2024). Given its high density of facilities, Virginia accounts for a disproportionate share of this load growth.

Despite this rapid expansion, regulatory oversight of large data center loads remains decentralized and evolving. There are no dedicated federal or PJM-level mandates governing the siting or cost allocation of large loads. At the state level, legislative efforts to require infrastructure assessments or address cost allocation have been proposed but not enacted (Virginia General Assembly, 2025; Virginia Mercury, 2025). As a result, regulation has largely been determined at the local level.

Local governments have historically permitted by-right development in many jurisdictions, though several counties have only recently introduced more restrictive zoning and review processes. For example, Loudoun County removed by-right zoning for data centers in 2025 (Loudoun County Government, 2025), Fairfax County adopted setback and design standards (Fairfax County Government, 2025), and other counties implemented buffer requirements (Henrico County Government, 2025) and infrastructure capacity reviews (York County Government, 2025). These measures vary across jurisdictions, reflecting a fragmented regulatory landscape.

Federal and regional discussions have acknowledged reliability and cost-allocation concerns associated with large loads. The Federal Energy Regulatory Commission convened a technical conference in 2024 (FERC, 2024), and the North American Electric Reliability Corporation established a Large Loads Task Force (NERC, 2025). However, no uniform regulatory framework has emerged. Consequently, Virginia provides a setting in which rapid, geographically concentrated demand growth has occurred under decentralized and recently evolving oversight.

2.2 Electricity Market Design and Nodal Pricing

In wholesale electricity markets, prices are determined using LMP. LMPs measure the marginal cost of delivering electricity at specific points on the transmission network, reflecting generation costs, local transmission congestion, and losses. Because they incorporate physical network constraints, LMPs provide granular and location-specific price signals that capture spatial variation in electricity market conditions.

LMPs are calculated at pricing nodes distributed throughout the transmission system. A pricing node represents a physical or operational point where electricity is injected or withdrawn, such as substations, generation injection points, or major load centers. In addition to these physical locations, system operators may define pricing nodes at hypothetical or mathematical points within the network to improve price granularity and transparency. These nodes collectively determine the geographic resolution at which wholesale prices vary.

Importantly, the configuration of pricing nodes is not static. System operators periodically create new pricing nodes in response to significant changes in the network, including the introduction of large loads or new transmission infrastructure. The interconnection of data centers frequently triggers the creation of new pricing nodes, as their substantial and persistent demand requires more localized price signals to manage congestion and maintain reliability.

This dynamic node creation poses a challenge for empirical analysis. A node-level panel

would be unbalanced, as new pricing nodes are introduced precisely in locations experiencing demand shocks. These newly created nodes do not have observations during the pre-treatment period, making it difficult to compare treated and untreated nodes prior to data center expansion. Appendix Figure A.1a illustrates this issue by distinguishing pricing nodes that exist throughout the entire sample period (Balanced pricing nodes) from the full set of nodes observed during 2015-2024 (All pricing nodes). The figure shows moderate entry of new nodes over time, highlighting the dynamic nature of node creation. Appendix Figure A.1b overlays node locations with data center sites and indicates that some newly created nodes are co-located with data centers. Consistent with this pattern, Appendix Table A.1 shows that pricing node creation is positively and statistically significantly associated with data center presence. Together, these patterns suggest that node entry is systematically related to data center expansion, complicating node-level analysis.

To address this challenge, we aggregate nodal LMP data to the census tract level. This approach offers two advantages. First, census tract boundaries remain fixed throughout our study period (2015-2024), allowing us to construct a balanced panel that is not affected by endogenous node entry. Second, aggregation ensures that tract-level prices incorporate newly created pricing nodes. When a new node is introduced following data center interconnection, its LMP is included in the tract-level measure, allowing us to capture localized price effects as the grid operator refines congestion signals in response to concentrated demand. For tracts containing multiple pricing nodes, we compute the average of nodal LMPs to construct tract-level prices.

2.3 Supply Constraints and Other Demand Conditions

Generation supply in Virginia is constrained by substantial interconnection backlogs during our study period. According to Rand et al. (2024), the average time from project proposal to commercial operation increased from less than two years in 2000-2007 to more than four years in 2018-2023, with PJM experiencing one of the most severe backlogs among U.S. wholesale electricity markets. These extended development timelines limit short-run supply responses to localized demand growth.

Consistent with this institutional environment, Appendix Figure A.2a shows that total generation capacity in Virginia remains relatively stable between 2015 and 2024. The absence of substantial capacity additions suggests that generation supply does not adjust contemporaneously to data center expansion.

Over the same period, Virginia did not experience other large-scale demand shocks comparable in magnitude or geographic concentration to data center entry. One potential con-

cern is the concurrent entry of energy-intensive manufacturing facilities in similar locations. However, Appendix Figure A.2b shows that the number of manufacturing plants remains largely stable throughout the sample period, suggesting no substantial influx of other major electricity-intensive demand sources.

Section 6.4 presents robustness checks that control for observed supply and demand factors; the results remain qualitatively similar to the main estimates.

3 Theoretical Framework

To analyze the impact of data center expansion on wholesale electricity prices, we use a merit-order framework to distinguish the underlying channels through which load growth affects LMPs.

LMP consists of three components: the system energy price, the congestion price, and the marginal loss price. The system energy price reflects the marginal cost of supplying one additional MWh in the absence of transmission constraints or losses and is uniform across all nodes within a market at a given point in time. The congestion price captures deviations arising from binding transmission constraints, which prevent electricity from flowing along least-cost paths and cause prices to diverge across locations. The marginal loss price reflects incremental transmission losses, as delivering one MWh of electricity to a given node requires generating more than one MWh due to line resistance. Unlike the system energy price, congestion and marginal loss prices vary spatially.

Formally, the LMP at node i and time t can be written as

$$\text{LMP}_{it} = \underbrace{\text{System Energy Price}_{t}}_{\text{uniform across nodes}} + \underbrace{\text{Congestion Price}_{it}}_{\text{transmission constraints}} + \underbrace{\text{Marginal Loss Price}_{it}}_{\text{line losses}}. \tag{1}$$

This decomposition clarifies how concentrated demand growth affects prices. At the system level, an increase in total load may raise the system energy price if higher-cost generators are dispatched at the margin, affecting all nodes uniformly (Figure 1a). At the local level, when transmission constraints bind, additional demand at a specific location cannot be met by increased power flows from lower-cost generators elsewhere. In this case, locational prices rise through higher congestion prices (Figure 1b). Data centers may also affect the marginal loss component. Higher demand increases total transmission flows, raising resistive losses. Moreover, if incremental load must be met by generators located farther away, longer transmission distances further increase losses (Figure 1c).

Although congestion and marginal loss prices arise from different physical mechanisms, both reflect deviations from the uniform system energy price driven by transmission constraints and delivery conditions. In our main results, we therefore focus on their combined

effect, which we refer to as the transmission-induced price component. This captures the spatial component of LMP variation induced by infrastructure constraints.

This framework motivates our empirical strategy. Because the system energy price is uniform across nodes at a given time, data center entry should not generate tract-specific variation in this component. Instead, localized effects operate through the congestion and marginal loss components, which vary spatially. We therefore estimate tract-level changes in these spatial price components to isolate transmission-related price effects associated with concentrated load growth.

4 Data

This paper relies on several data sources. First, we obtain real-time hourly LMP data from PJM Data Miner 2 for the period 2015-2024. PJM reports LMPs and their decomposition into system energy, congestion, and marginal loss components. Consistent with the conceptual framework, we focus on the combined congestion and marginal loss components as our primary outcome, which capture transmission-related deviations from the uniform system energy price. Because pricing nodes are created dynamically over time, we aggregate nodal LMPs to the census tract level to construct a tract-year panel. For each tract, we also compute annual averages of hourly prices and construct season- and hour-specific averages to examine heterogeneity across time periods.

Second, we obtain data center location and operation-year information from the Data Center Map.[2] We aggregate data center entries to the census tract level and define the treatment year as the first year in which a data center becomes operational in a tract. Our main analysis is restricted to tracts whose first data center entry occurs in 2017 or later, ensuring sufficient pre-treatment observations in the panel. Tracts with data center operations prior to 2017 are excluded from the sample.

Third, we collect generation capacity and plant location data from the U.S. EIA Form 860. To assess other potential demand shocks, we collect manufacturing facility data from the U.S. EPA Toxic Release Inventory (TRI).[3] Finally, we obtain tract-level demographic characteristics from the American Community Survey (ACS), which we use for matching and distributional analysis.

Figure 2 maps the locations of all pricing nodes that appear in Virginia together with data centers entering between 2015 and 2024. Pricing nodes are distributed across the state,

[2]Operation years are sparsely recorded in the raw dataset. We supplement missing values by manually collecting operation dates from publicly available news sources and cross-validating across outlets, including public announcements.

[3]Facilities are included in TRI if they exceed employment thresholds and operate within covered industrial sectors, including manufacturing (EPA, 2025).

providing broad geographic coverage for constructing tract-level price measures. Data centers are concentrated primarily in Northern Virginia, where pricing nodes are also more densely located. Tracts without data center entry during the sample period serve as untreated tracts. Figure 3 shows the distribution of first data center entry years for tracts included in the analysis sample.

Figure 4 reports the distribution of distances between the treated census tracts and the nearest generation facility in 2015. Fewer than five of the nineteen treated census tracts in our analysis sample have a generation facility within a five-kilometer radius. Most generation facilities are located more than fifteen kilometers away from the treated tracts. Given the geographic scale of census tracts in Virginia, these distances indicate that nearby generation is often located further away rather than within the same tract as a data center.

Appendix Figure A.2 summarizes supply- and demand-side conditions over time. Total generation capacity in Virginia remains relatively stable throughout the sample period. The number of TRI manufacturing facilities also changes little over time, suggesting limited contemporaneous supply expansion or other large demand shocks.

Table 1 reports summary statistics for census tracts with and without data center entry. Panels A and B compare pre-treatment demographic characteristics in 2015-2016 across treated and untreated tracts. Treated tracts are more densely populated and have higher income and educational attainment on average. Panel C reports LMPs and their components, which also differ across groups on average. The observed differences in both tract characteristics and outcomes motivate our two-stage empirical strategy, which combines propensity score matching with difference-in-differences estimation.

5 Empirical Strategy

5.1 First Stage: Propensity Score Matching

Because treated and untreated tracts differ systematically in observable characteristics, we first construct a matched sample based on pre-treatment demographics. We estimate the probability that a census tract experiences its first data center entry between 2017 and 2024 using a probit model (Caliendo and Kopeinig, 2008):

$$Pr(DC_i = 1|X_i) = \Phi(X_i'\gamma), \tag{2}$$

where DC_i is an indicator equal to one if tract i experiences data center entry during the sample period. The vector X_i includes tract-level characteristics measured in 2015-2016, prior to treatment, including population, percentage of White residents, percentage of high school graduates, median household income, and number of housing units.

Appendix Figure A.3a shows the distribution of estimated propensity scores prior to matching. The horizontal axis reports the propensity score, and the vertical axis shows the share of census tracts within each bin. Adequate common support requires similar distributions across treated and untreated tracts. As shown in the figure, both groups present right-skewed distributions, though untreated tracts are considerably more concentrated at very low propensity scores. In addition, one treated tract has a propensity score that exceeds the support of the untreated group.

Using the estimated propensity scores, we implement $k-$nearest neighbor (kNN) matching with replacement, matching each treated tract to its ten nearest untreated neighbors (10NN). This process pairs treated tracts with similar untreated tracts based on observed pre-treatment characteristics, restricting the sample to comparable units.

Appendix Figure A.3b presents the distribution of propensity scores after matching. The distributions of treated and untreated tracts become substantially more comparable, and the treated tract outside common support is excluded from the matched sample.[4]

Table 1 columns 3-4 report the summary statistics of census tracts with and without data centers after matching. Differences in demographic characteristics are substantially reduced and no longer statistically significant after matching. In contrast, differences in LMPs persist: treated tracts have higher average transmission-induced price ($5.34 vs. $3.62 per MWh).

5.2 Second Stage: Staggered Difference-in-Differences Models

In the second stage, we estimate the impact of data center entry on transmission-related price component using a staggered difference-in-differences design. The specification is:

$$Y_{it} = \alpha + \beta DC_{it} + \gamma_i + \omega_t + \epsilon_{it}, \tag{3}$$

where Y_{it} denotes the transmission-induced price component, defined as the combination of congestion and marginal loss components of LMP in census tract i and year t. The indicator DC_{it} equals one for tract i in all years t at or after its first data center entry and zero otherwise. The model includes tract fixed effects γ_i and year fixed effects ω_t. Because propensity score matching is conducted with replacement, some untreated census tracts may appear multiple times in the matched sample. To account for this, we weight each observation by the inverse of the number of times a tract appears in the matched dataset. Identification relies on variation in the congestion and marginal loss components, which reflect spatial

[4] Appendix Figure A.4 shows the locations of the treated and untreated census tracts after matching.

price differences associated with transmission constraints. Standard errors are clustered at the census tract level.

Event Study Specification

To assess the parallel trends assumption and examine dynamic treatment effects, we estimate an event study specification:

$$Y_{it} = \Big(\sum_{\tau \in \{-q,\ldots,-2,0,\ldots,+p\}} \delta_\tau \cdot DC_{i,t-\tau} \Big) + \gamma_i + \omega_t + \epsilon_{it} \tag{4}$$

The summation allows for q leads $(\delta_{-2}, \ldots, \delta_{-q})$ and p lags $(\delta_0, \ldots, \delta_{+p})$, where $DC_{i,t-\tau}$ is an indicator for event time τ. It equals one if calendar year t is τ years before census tract i introduces its first data center. The δ_τ terms for $\tau \geq 0$ capture the dynamic impact of the treatment over time since the first year of data center entry. The δ_τ terms for $\tau < 0$ provide a placebo test, showing the treatment effect before the event occurs. If no confounding variables are omitted, these pre-event terms (δ_τ for $\tau < 0$) should be statistically insignificant from zero. In other words, we should not observe a trend during the pre-treatment period.

Estimation Approach

Our preferred estimator follows Sun and Abraham (2021), which addresses the negative weighting problem that can arise in standard two-way fixed effects (TWFE) models under staggered treatment timing (Goodman-Bacon, 2021). For robustness, we also estimate the model using the TWFE specification and the approach of Callaway and Sant'Anna (2021). The estimated effects are qualitatively similar across methods. This consistency is expected given the presence of a large never-treated group, such that most of the identifying variation comes from comparisons between treated and untreated tracts.

Identification relies on two key assumptions. First, we assume parallel trends: in the absence of data center entry, transmission-induced price component in treated and untreated tracts would have evolved along similar trajectories. We provide evidence on this assumption through pre-treatment event-study coefficients in Section 6.1. Second, we assume no anticipation: outcomes do not respond to data center entry before the facility becomes operational. Although data centers involve long planning and permitting timelines, grid-level impacts- particularly congestion and marginal loss effects, are driven by actual electricity consumption rather than the announcement or construction of a facility. We therefore define treatment as the year a data center begins drawing power from the grid.

We note one additional consideration. Our staggered difference-in-differences design as-

sumes that the Stable Unit Treatment Value Assumption (SUTVA) holds — specifically, that data center entry in one tract does not affect outcomes in other tracts. In a meshed transmission network, congestion at one node can affect flows and prices at other nodes, potentially violating this assumption. We examine this concern directly in Section 6.4.

6 Results

6.1 Average Impact of Data Center Entry on Transmission-Induced Price Component

Figure 5 presents event-study estimates of the impact of data center entry on transmission-induced price component, defined as the combined congestion and marginal loss components of LMP. The horizontal axis reports event time relative to the year of first data center entry in a census tract, and the vertical axis shows changes in transmission-induced price component in each year relative to the year prior to entry.

The transmission-induced component begins to increase in the second year after entry and stabilizes by the third year. Although individual post-treatment coefficients are not statistically significant at the 5 percent level, the estimate in year five is significant at the 10 percent level. Pre-treatment coefficients are small and statistically insignificant at the 10 percent level, providing support for the parallel trends assumption. Aggregating post-treatment effects yields an average increase of $2.49 per MWh (Table 2 column 1). This estimate is statistically significant and corresponds to a 70 percent increase relative to pre-treatment levels.[5]

Figure 6 decomposes the effect into congestion and marginal loss components. Both components increase following data center entry, with effects emerging in the second year. The marginal loss component is statistically significant at the 5 percent level in later years. Aggregating post-treatment effects, data center entry increases congestion prices by $2.32 per MWh and marginal loss prices by $0.17 per MWh (Table 2, columns 2-3). Pre-treatment coefficients remain small and statistically insignificant for both components.

Appendix Figure A.6 reports event-study estimates for system energy price and total LMP. As expected, system energy price shows no differential effect following data center entry, consistent with its uniform determination across nodes within the same market. On the other hand, total LMP closely mirrors the pattern observed for the transmission-induced price component, reflecting that variation in total LMP is driven by congestion and marginal

[5]Appendix Figure A.5 compares the aggregate effects on transmission-induced price component across the standard two-way fixed effects estimator, the estimator of Callaway and Sant'Anna (2021), and our preferred estimator of Sun and Abraham (2021). The results are qualitatively similar across approaches.

loss components.[6] The null effect on the system energy price serves as a useful placebo: because this component is determined uniformly across the market, the entry of a data center should not generate a differential system energy effect, and the absence of such an effect is consistent with our identification strategy.

6.2 Mechanism: Generation Proximity and Fuel Type

To investigate mechanisms underlying the increase in the transmission-induced price component, we examine the geographic proximity between the treated census tracts and generation resources.

Table 3 column 1 reports heterogeneity by the distance between a data center tract and its nearest generation resource. Distance is measured in kilometers. The interaction between data center entry and distance is positive and statistically significant, indicating that the effect on the transmission-induced price component is larger in tracts where generation is located farther away. The coefficient on the data center indicator itself is small and statistically insignificant, suggesting that when generation is located very close to the data center tracts, the estimated effect is close to zero.[7]

We next examine heterogeneity by fuel type for generation located within the same census tract as the data center. Table 3 column 2 reports interactions between data center entry and tract-level generation capacity by fuel type. Fossil fuel capacity is a continuous variable measuring average generation capacity across years from coal, natural gas, and oil - resources typically considered dispatchable. Renewable capacity measures average generation capacity from solar, wind, and biofuel, which are generally treated as intermittent, non-dispatchable resources. Together, fossil and renewable resources account for nearly 95 percent of total generation capacity across census tracts in our sample.

The interaction between data center entry and fossil fuel capacity is negative and statistically significant, indicating that transmission-induced price component is lower following data center entry when data centers are co-located with greater fossil fuel capacity within the same tract. In contrast, the interaction with renewable capacity is positive and statistically significant, indicating that prices are higher when data centers are co-located with greater renewable capacity. These results indicate that the estimated price response to data center entry differs systematically by the fuel type of co-located generation. Dispatchable fossil fuel generation can respond directly to increased local load, limiting transmission-related spa-

[6] Appendix Table A.2 reports the corresponding aggregate estimates for total LMP and the system energy price. Appendix Figure A.7 compares the aggregate effects on different components of LMP separately across the standard two-way fixed effects estimator, the estimator of Callaway and Sant'Anna (2021), and our preferred estimator of Sun and Abraham (2021). The results are consistent across specifications.

[7] Among the nearest generation resources, 74 percent are fossil-fuel based and 19 percent are renewable.

tial price wedges. Renewable generation, by contrast, is intermittent, so additional demand may still need to be met by importing dispatchable resources from more distant locations, increasing congestion and marginal loss components even when renewable capacity is nearby.

Lastly, we explore heterogeneity by fuel type of the nearest generation resource, rather than restricting to generation located within the same census tract as the data center. Table 3 column 3 reports these results. The interaction terms between data center entry and fuel type are statistically insignificant for both fossil fuel and renewable generation. This indicates that when generation is not co-located within the same census tract as the data center, the estimated price response does not vary systematically across fuel types.[8]

Taken together, these findings provide suggestive evidence that increases in transmission-induced price component are driven by the spatial misalignment between data center load and generation capacity. The estimated effects are larger when generation is located farther away, and fuel-type heterogeneity arises only when generation is co-located within the same census tract as the data center. In such cases, dispatchable fossil fuel capacity is associated with smaller price responses, whereas renewable capacity is associated with larger responses.

6.3 Heterogeneity

We examine heterogeneity along several dimensions, including temporal variation, treatment intensity, and demographics.

6.3.1 Temporal Variation

We examine heterogeneity across hours of the day and seasons to assess whether the effects of data center entry are concentrated during peak demand periods.

We first aggregate the transmission-induced price component to the tract-hour-year level and estimate the post-treatment effect separately for each hour of the day. Figure 7a reports the aggregate effect for each hour. Each dot represents the point estimate from a separate regression, and the bars denote 95 percent confidence intervals.

The results show pronounced hourly heterogeneity. Effects are small during overnight off-peak hours (approximately 12am–6am) and increase more substantially during afternoon peak hours, particularly around 4–5pm. Morning peak hours beginning around 7am also show positive effects, though smaller in magnitude than the afternoon peak. After 5pm, estimated effects decline but remain positive. This pattern indicates that transmission-

[8]Appendix Table A.3 replicates the three heterogeneity analyses using congestion and marginal loss components separately as outcome variables. The results are qualitatively similar to those obtained using the combined transmission-induced price measure.

induced price effects associated with data center entry are concentrated during periods of elevated system demand.

We next examine seasonal heterogeneity by aggregating the transmission-induced price component to the tract-season-year level. We define March-May as Spring, June-August as Summer, September-November as Fall, and December-February as Winter. Figure 7b presents aggregate effects by season. The estimated effects are broadly similar across seasons, with slightly larger magnitudes in Spring.

To further assess peak dynamics, we estimate effects separately by hour within each season. Figure 8 shows that the hourly pattern observed in the aggregate results holds across different seasons. Effects are larger during afternoon peak hours, a pattern that holds for Spring, Summer, and Fall. In contrast, winter exhibits relatively limited hourly heterogeneity. This pattern may be driven by the lower reliance on electricity for heating in Virginia, where only approximately 60 percent of households use electricity as their primary heating source (US EIA, 2025), which may dampen winter peak sensitivity relative to other seasons.[9]

Finally, we estimate dynamic event-study specifications separately by hour and season. Appendix Figures A.11 and A.12 show post-treatment increases in congestion and marginal loss prices that are more pronounced during afternoon and morning peak hours. Comparing seasons, summer exhibits larger dynamic effects than winter, consistent with seasonal demand variation.

6.3.2 Treatment Intensity

We next investigate heterogeneity by treatment intensity. Table 4 column 1 reports the effect of additional data center entries within a census tract, where the number of data centers is measured as a continuous variable capturing the cumulative count of data centers operating in a tract in each year. The results indicate that the transmission-induced price component increases with treatment intensity: each additional data center is associated with an increase of $0.81 per MWh. Appendix Table A.4 reports the corresponding estimates separately for the congestion and marginal loss components, both of which show statistically significant increases.

These findings relate to the discussion in Borenstein (2025), which suggests that moderate increases in data center capacity may *reduce retail* electricity rates by spreading network

[9]We replicate these exercises separately for congestion and marginal loss components of LMP. Appendix Figures A.8, A.9, and A.10 report the corresponding results. The patterns are qualitatively similar: congestion and marginal loss prices both exhibit stronger effects during afternoon peak hours and limited seasonal variation in winter.

(fixed) costs over higher consumption. In contrast, our findings speak to the variable-cost channel, where we find no systematic evidence that data center entry reduces the transmission-induced price component under current conditions. Instead, greater treatment intensity is associated with higher transmission-induced price effects.[10]

6.3.3 Demographics

We also examine heterogeneity across demographic characteristics to assess distributional patterns. Table 4 columns 2-4 present interactions between data center entry and tract-level population, median household income, and the percentage of White residents.[11]

The data center indicator remains statistically significant across specifications. The interaction terms with population and household income are statistically insignificant, indicating no systematic heterogeneity along these dimensions. In contrast, the interaction between data center entry and the percentage of White residents is negative and statistically significant. This implies that the estimated increase in the transmission-induced price component following data center entry is larger in tracts with lower shares of White residents, that is, in predominantly non-White communities.[12] These findings indicate that the transmission-induced spatial price wedges associated with data center entry vary systematically by racial composition, but not by income or population size.

6.4 Robustness

We address potential concerns through a series of robustness checks. In particular, we examine the sensitivity of our results to identification assumptions and to the matching procedure.

6.4.1 Identification

Supply and Other Demand Shocks

A key identification assumption is that the observed price changes are primarily driven by demand-side shocks associated with data center entry rather than contemporaneous supply

[10] Our analysis focuses on wholesale electricity price changes induced by transmission constraints and uses the number of data centers rather than load capacity due to data limitations. Although wholesale prices do not map one-for-one into retail rates, they provide an informative indicator of underlying pricing pressures faced by downstream markets.

[11] We use demographic information from 2015, during the pre-treatment period, to define heterogeneous groups.

[12] Appendix Table A.5 reports the corresponding heterogeneity estimates separately for congestion and marginal loss components. The results are qualitatively similar to those in Table 4 columns 2-4.

expansions. We support this assumption by referencing institutional evidence on supply constraints in Virginia. During our study period, new generation faced substantial permitting and construction lead times, as well as severe interconnection delays within PJM (Gorman et al., 2025), both of which limited short-run supply responses.[13]

As an additional robustness check, Appendix Table A.6 columns 1-5 re-estimate the aggregate effects of data center entry on LMP components while controlling for total generation capacity to account for gradual supply changes. The estimates remain qualitatively similar to those reported in Table 2 and Appendix Table A.2.

On the demand side, we address the possibility that the observed variation reflects broader demand growth rather than shocks specifically associated with data center entry. One potential confounder is the entry of energy-intensive manufacturing facilities. Appendix Table A.6 columns 6–10 include controls for the number of manufacturing plants. The estimated effects of data center entry remain stable and statistically similar to the baseline results.

Spillovers of Transmission-induced Price

Another concern is that congestion induced by data center entry may spill over to nearby tracts. In a meshed transmission network, congestion at one location can alter power flows and prices at distant nodes through loop flows, potentially violating SUTVA. If untreated tracts are affected by congestion originating in treated tracts, the control group outcomes are contaminated, biasing the estimated effect toward zero (if spillovers raise control group prices) or away from zero (if spillovers lower control group prices through rerouting).

To examine this possibility, we compute pairwise correlations in the pre-treatment transmission-induced price component across all census tracts in Virginia. We identify untreated tracts that are highly correlated with treated tracts and exclude untreated tracts with correlation coefficients greater than 0.5 relative to treated tracts. This procedure restricts the comparison group to tracts that are less likely to be affected by spillovers through the transmission network.

The restricted sample includes 20 treated tracts and 30 untreated tracts. Appendix Table A.7 Panel A reports estimates using this lower-correlation untreated group. The results are consistent with the baseline aggregate estimates, suggesting that spillovers are unlikely to drive the main findings.[14]

[13] We also verify that there was limited expansion of transmission lines in Virginia during our study period; over 80 percent of new transmission capacity was added after 2025 (VA State Corporation Commission, 2025).

[14] The choice of the 0.5 correlation threshold is somewhat arbitrary. In unreported results, we estimate the model using alternative thresholds ranging from 0.5 to 0.7 and obtain qualitatively similar results.

6.4.2 Matching

Our results are robust to alternative matching procedures. Appendix Table A.7 reports estimates from several alternative specifications: models without propensity score matching (Panel B), models not weighted by tract duplication arising from matching with replacement (Panel C), models excluding duplicated tracts entirely (Panel D), and models using five nearest neighbors (5NN) instead of ten (10NN) for matching (Panel E). The estimated effects remain stable across specifications, indicating that the findings are not sensitive to the matching design.

7 Conclusion

This paper estimates the causal effect of concentrated demand shocks on spatially differentiated wholesale electricity prices using the expansion of data centers in Virginia between 2015 and 2024. Leveraging staggered data center entry across census tracts and a setting characterized by limited short-run supply responses, we show that data center development increases the transmission-induced price component, defined as the congestion and marginal loss components of locational marginal prices, by $2.49 per MWh on average, corresponding to an approximately 70 percent increase relative to pre-treatment levels.

The evidence points to transmission-related mechanisms. The transmission-induced price component is larger when generation resources are located farther from data center load. Fuel-type heterogeneity emerges when generation is co-located within the same census tract: prices are lower when data centers are co-located with greater dispatchable fossil fuel capacity and higher when co-located with greater renewable capacity. When generation is located outside the tract, fuel type does not systematically alter the price response. These patterns suggest that spatial alignment between load and dispatchable capacity plays an important role in moderating localized spatial price wedges.

We also document substantial heterogeneity across time and space. Transmission-induced price effects are concentrated in afternoon peak hours and are more pronounced in spring months. Effects scale with treatment intensity, as additional data center entries within a tract are associated with further increases in the transmission-induced price component. Finally, these price effects are larger in predominantly non-White communities, indicating that concentrated load growth may amplify existing spatial disparities in electricity markets.

These findings have implications for infrastructure planning and market design. As electricity demand shifts due to technological change, including the expansion of data centers and other large loads, transmission constraints can generate localized spatial price wedges even when aggregate generation capacity appears sufficient. Our results highlight the importance of coordinating load siting with generation and transmission development to mitigate

transmission-induced price effects. More broadly, the distributional patterns we document suggest that the geographic incidence of large-scale electrification may have uneven effects across communities.

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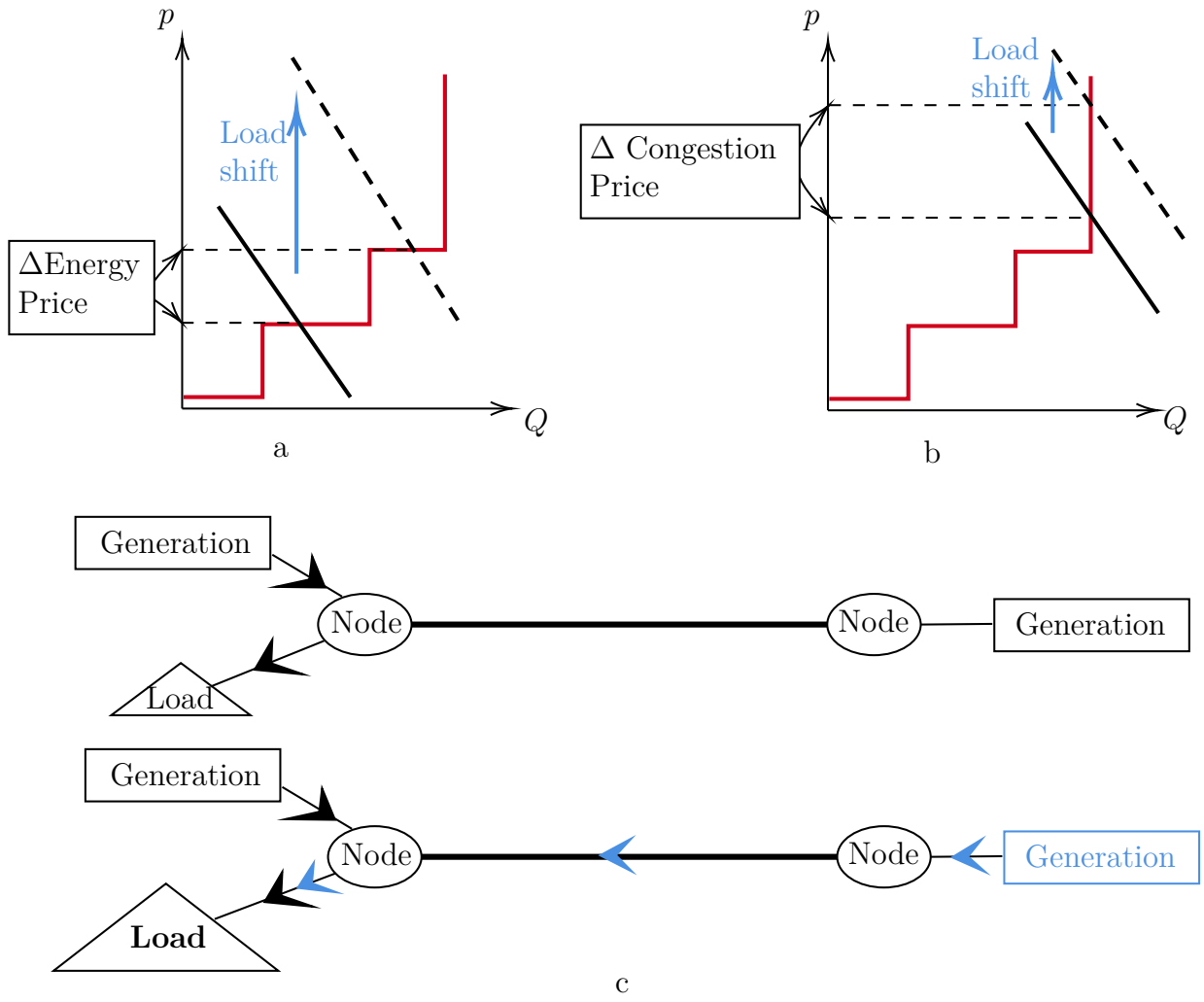
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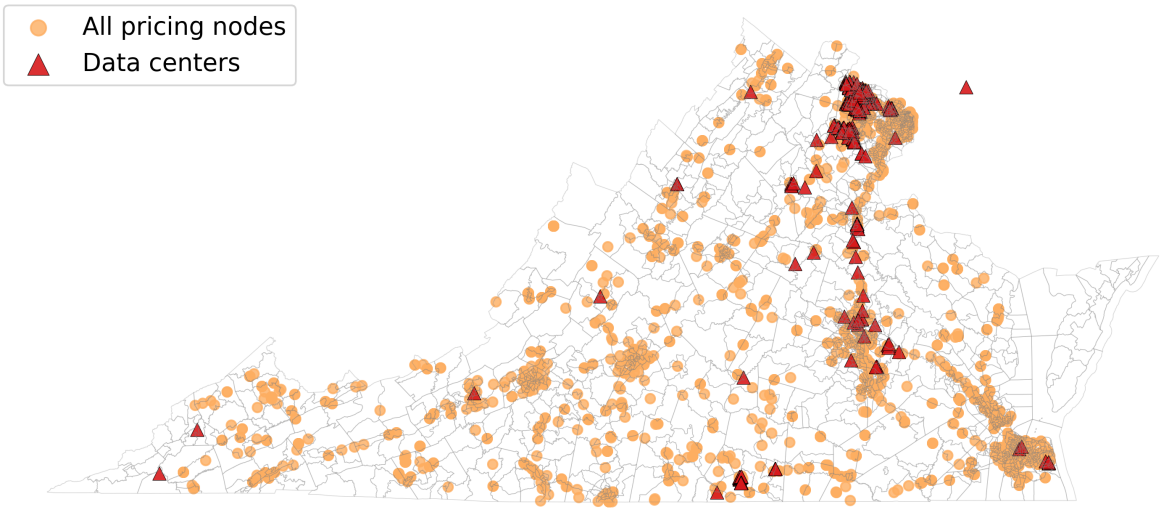
Figures and Tables

Figure 1: Demand Shift and Different LMP Components



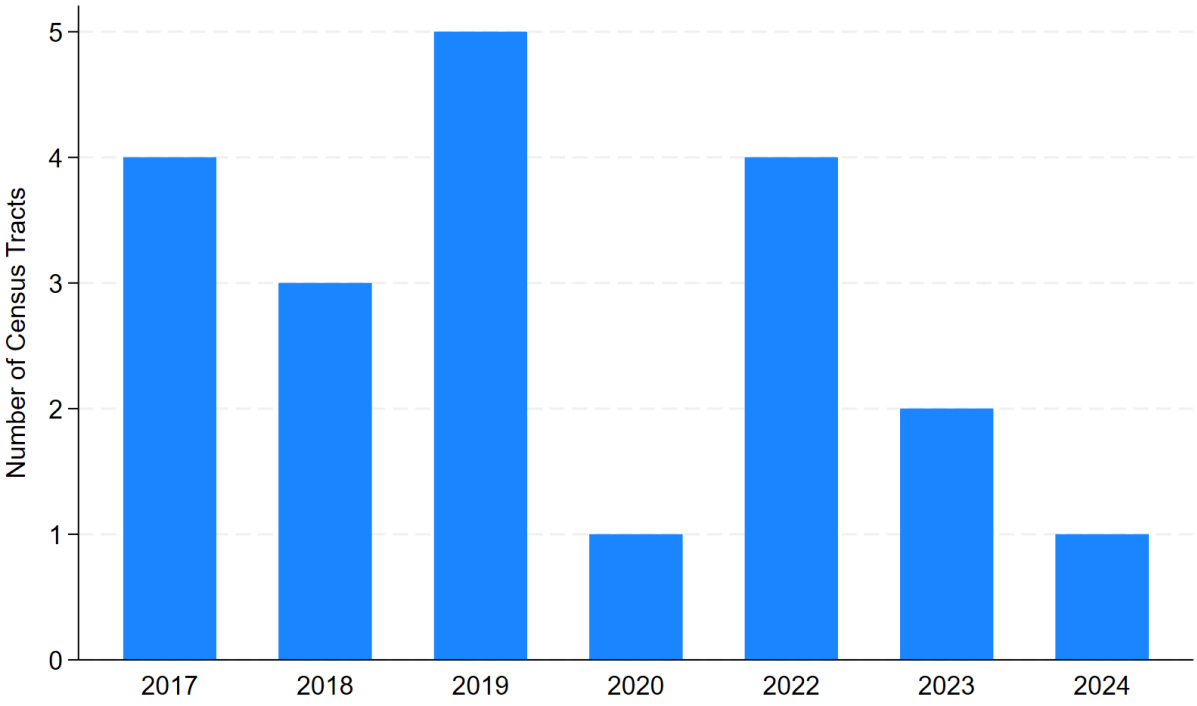
Notes: This figure illustrates how each component of locational marginal prices (LMPs) responds to an increase in load. Panel a depicts the system energy component, which rises when higher demand shifts the dispatch to generators with higher marginal costs along the merit order. Panel b illustrates the congestion component, which increases when additional load causes transmission constraints to bind, leading to spatial price divergence. Panel c shows the marginal loss component, which increases when serving additional demand requires power to be transmitted over longer distances, resulting in higher line losses.

Figure 2: Locations of Data Centers and All Pricing Nodes in Virginia



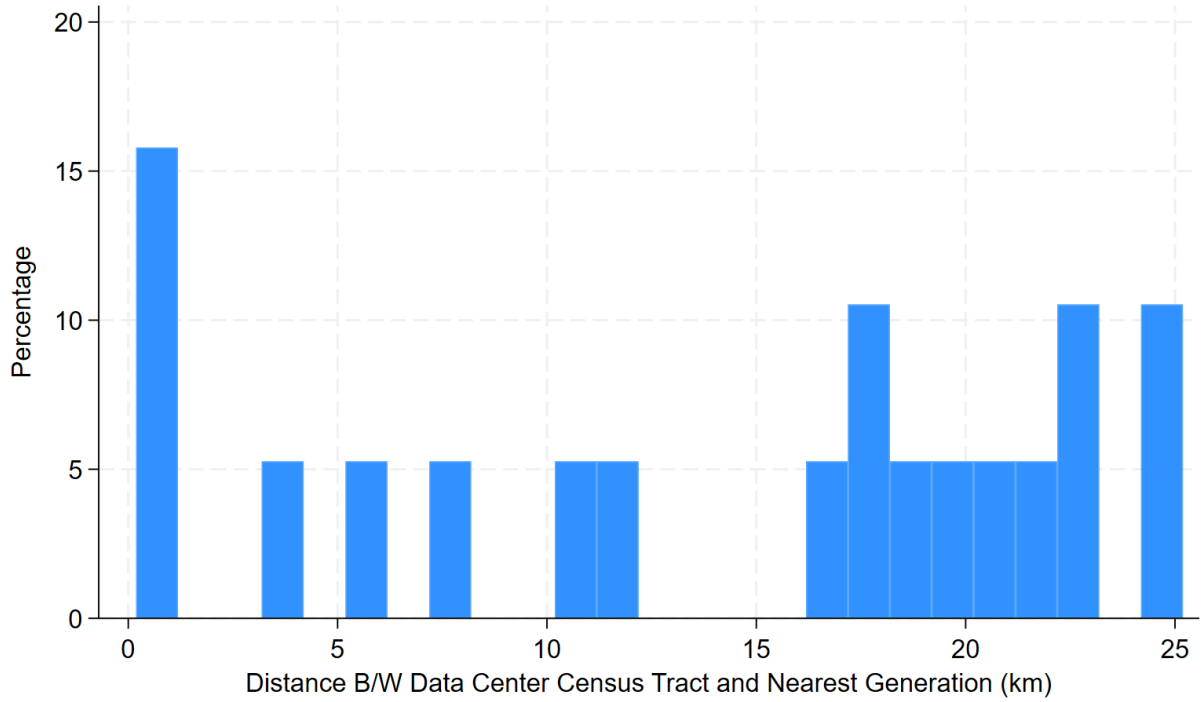
Notes: This figure presents a map of Virginia showing the locations of data centers and all pricing nodes that appear during the study period. Pricing nodes are distributed throughout the state. In areas with data center presence, pricing nodes are more densely concentrated.

Figure 3: Data Center Entry at the Census Tract Level in Virginia



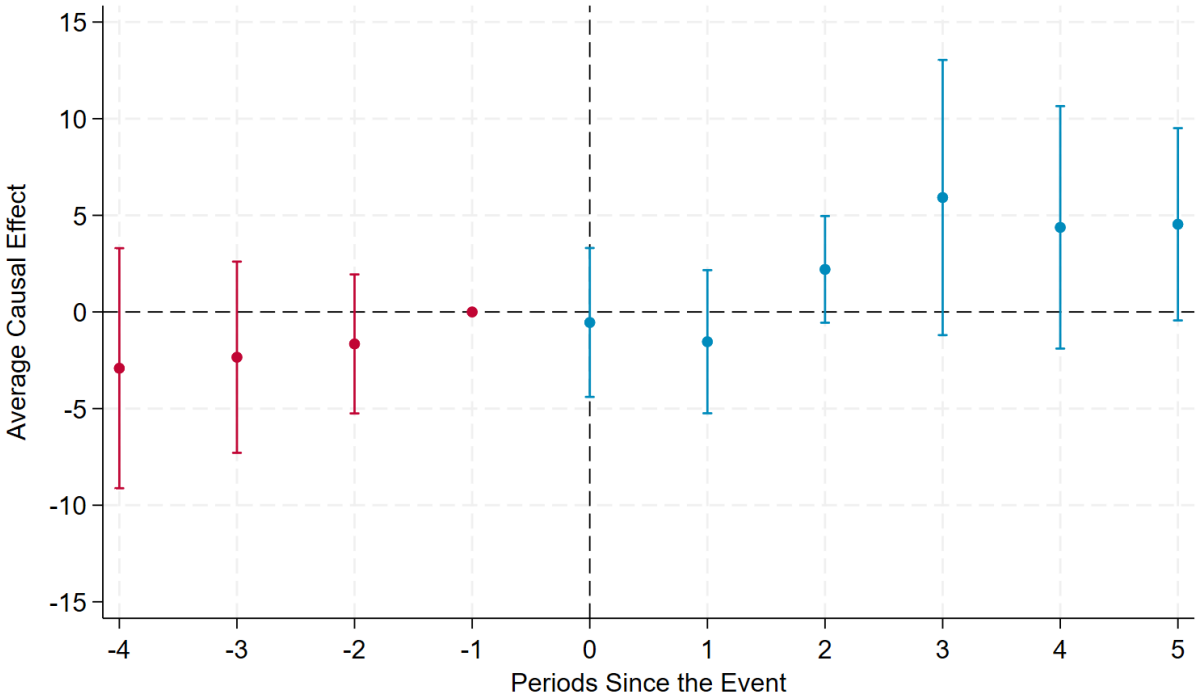
Notes: This figure reports the number of census tracts by year of first data center entry during the study period. For example, five census tracts experienced their first data center entry in 2017. The analysis sample is restricted to tracts with a first entry in 2017 or later.

Figure 4: Distance Between Data Center Census Tracts and the Nearest Generation Resource



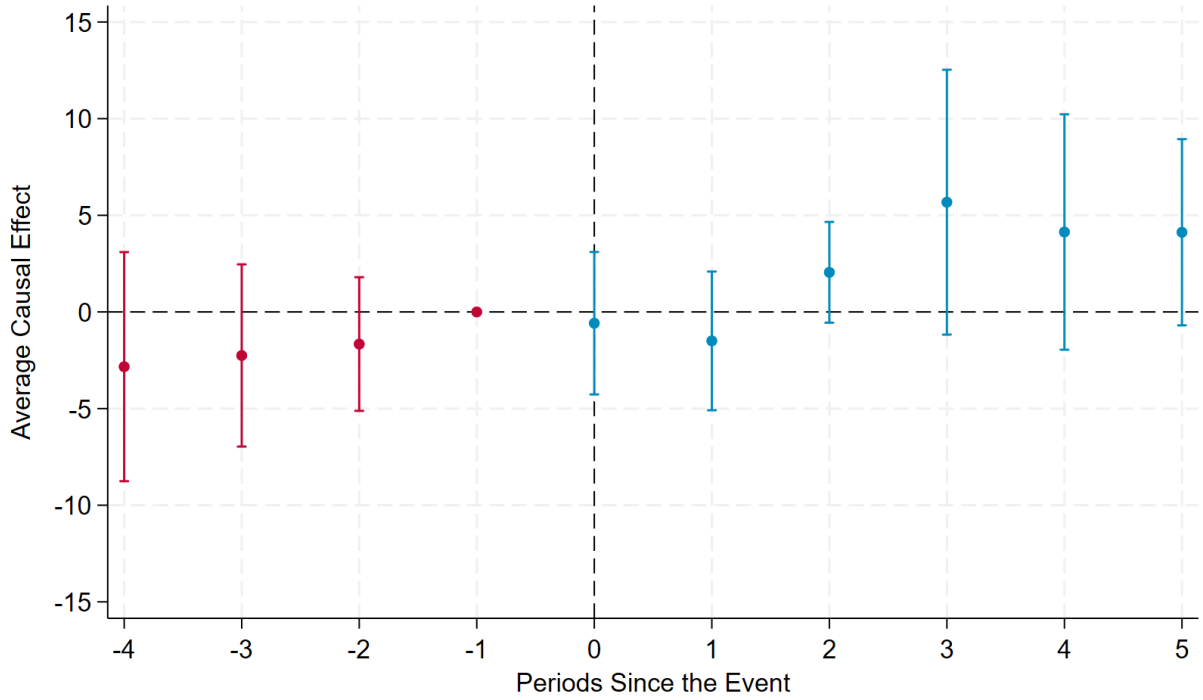
Notes: This figure presents the distribution of distances, measured in kilometers, between data center census tracts and their nearest generation resource. Distance is calculated at the pricing-node level as the distance between each pricing node within a census tract and its closest generation facility. For tracts with multiple pricing nodes, the tract-level distance is defined as the average of the nearest-generation distances across pricing nodes within the tract.

Figure 5: Dynamic Effects of Data Center Entry on Transmission-Induced Price Component

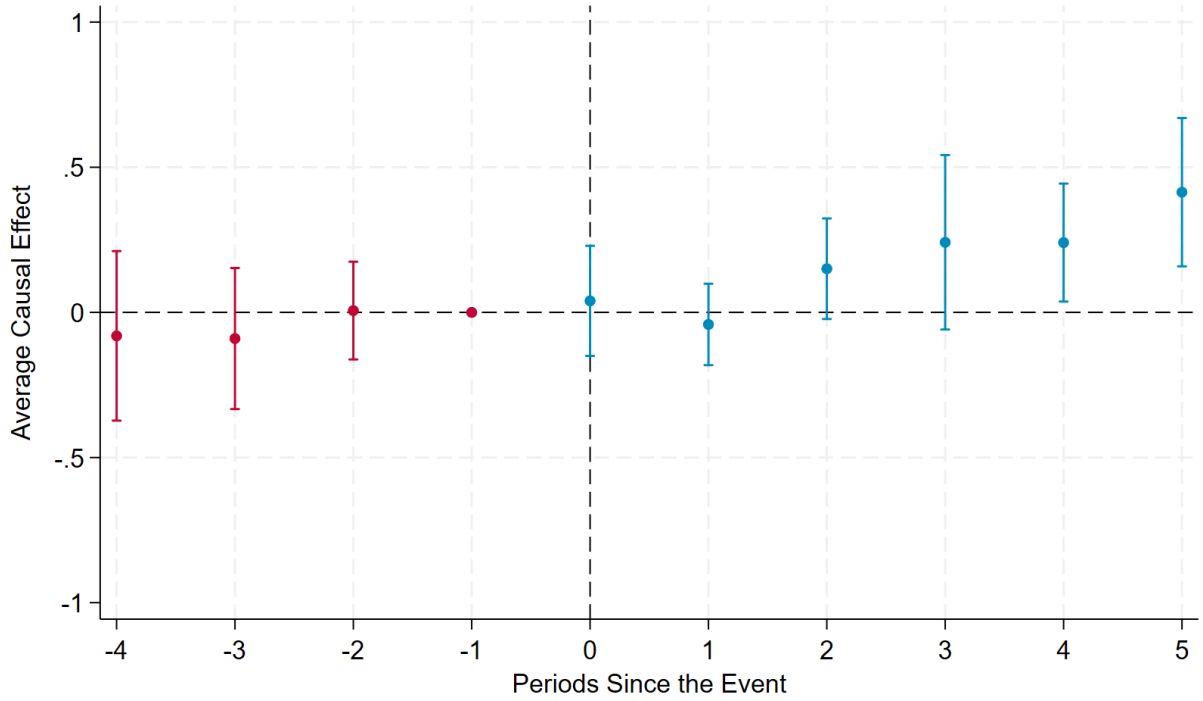


Notes: This figure presents event-study estimates of the effect of data center entry on the transmission-induced price component, defined as the sum of congestion and marginal loss components of LMP, using the estimator of Sun and Abraham (2021). Event times earlier than -4 and later than +5 are binned and not shown. The reference year is defined as one year prior to the first data center entry in a census tract. All regressions include census tract and year fixed effects. Because matching is conducted using ten nearest neighbors (10NN) with replacement, regressions are weighted by the inverse frequency of duplicated untreated tracts. Standard errors are clustered at the census tract level. Pre-treatment coefficients are small and statistically insignificant. Post-treatment coefficients increase beginning in year two and stabilize after year three. Although most individual post-treatment coefficients are not statistically significant at the 5 percent level, the year five estimate is significant at the 10 percent level. The aggregated post-treatment effect is statistically significant (Table 2 column 1).

Figure 6: Dynamic Effects of Data Center Entry on Congestion and Marginal Loss Price



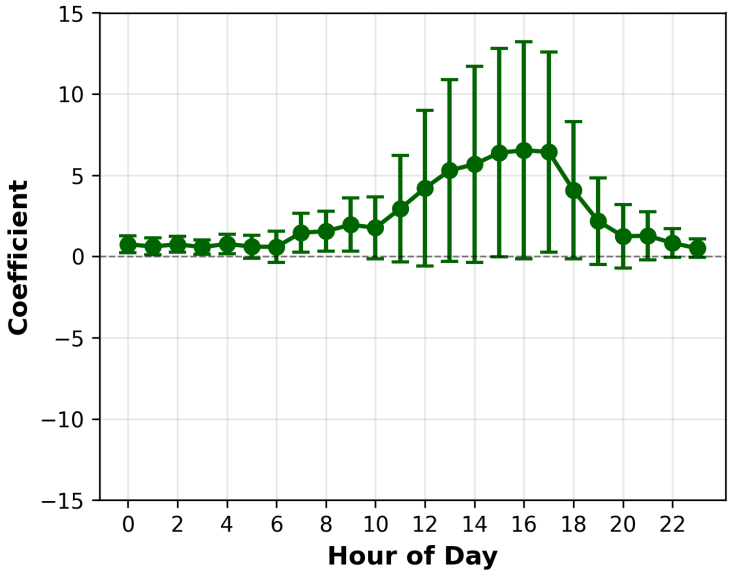
(a) Average Congestion Price



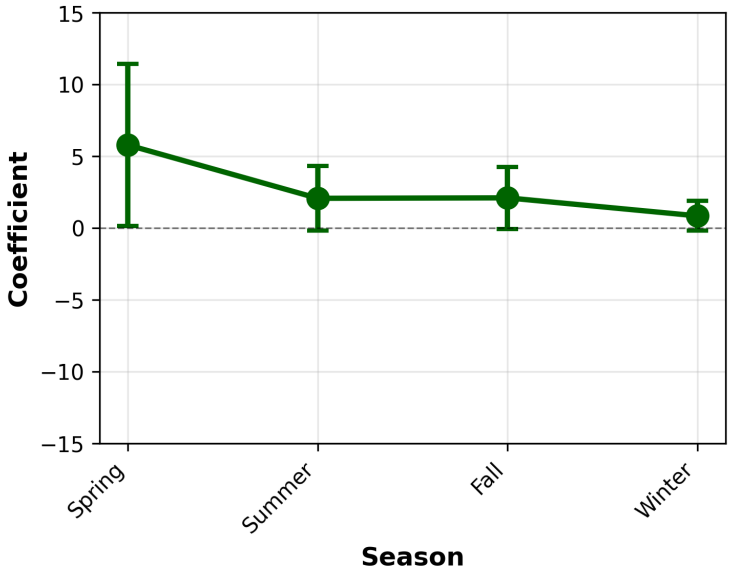
(b) Average Marginal Loss Price

Notes: This figure presents event-study estimates of the effect of data center entry on congestion price and marginal loss price, respectively, using the estimator of Sun and Abraham (2021). Event times earlier than -4 and later than +5 are binned and not shown. Standard errors are clustered at the census tract level. Pre-treatment coefficients are small and statistically insignificant in both panels. The aggregated post-treatment effects are statistically significant (Table 2 columns 2–3).

Figure 7: Impact of Data Center Entry on Transmission-induced Price Component by Hour and Season



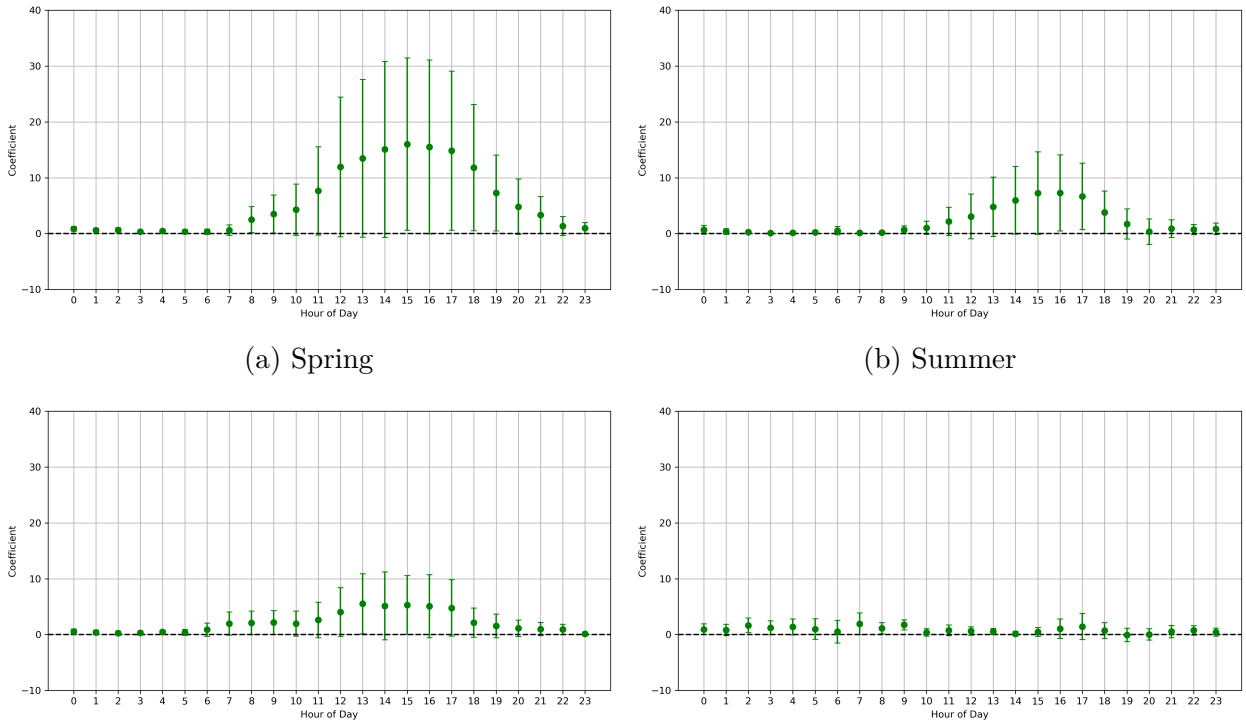
(a) Across Different Hours



(b) Across Different Seasons

Notes: This figure reports aggregate post-treatment effects of data center entry on the transmission-induced price component by hour (Panel a) and by season (Panel b). Each point represents the average treatment effect over the post-treatment period estimated separately for each hour or season. Bars denote 95 percent confidence intervals. Estimates are obtained using the estimator of Sun and Abraham (2021).

Figure 8: Impact of Data Center Entry on Transmission-induced Price Component by Hour Across Seasons



(a) Spring

(b) Summer

(c) Fall

(d) Winter

Notes: This figure reports aggregate post-treatment effects of data center entry on the transmission-induced price component by hour within each season. Each point represents the average treatment effect over the post-treatment period estimated separately for each hour–season combination. Bars denote 95 percent confidence intervals. Estimates are obtained using the estimator of Sun and Abraham (2021). Standard errors are clustered at the census tract level.

Table 1: Summary Statistics for Census Tracts Before and After Propensity Score Matching

	Before Propensity Score Matching			After Propensity Score Matching		
	Treated (1)	Untreated (2)	Diff (1) - (2)	Treated (3)	Untreated (4)	Diff (3) - (4)
Panel A: Demographic Characteristics in 2015						
Population	4,896.73	4020.05	876.68***	4,923.21	4,857.74	65.47
	[1,753.26]	[1,633.52]	(118.43)	[1,795.12]	[2,161.18]	(162.12)
Percent White	65.13	64.40	0.73	65.83	67.81	-1.98
	[19.11]	[25.89]	(1.85)	[19.35]	[20.77]	(1.57)
Percent High School Graduates	82.06	75.79	6.27***	82.93	82.67	0.26
	[16.23]	[21.12]	(1.51)	[16.20]	[19.94]	(1.49)
Median Household Income	89,370.80	54,646.06	34,724.74***	87,628.12	86,982.36	645.76
	[39,533.76]	[33,941.50]	(2,471.40)	[39,806.43]	[51,258.41]	(3,829.48)
Housing Units	1,781.96	1,718.47	63.49	1,807.51	1,759.91	47.60
	[492.77]	[639.50]	(45.82)	[492.48]	[587.65]	(44.11)
Panel B: Demographic Characteristics in 2016						
Population	5,025.48	4,044.10	981.38***	5,018.16	4,941.91	76.25
	[1,792.51]	[1,652.38]	(119.86)	[1,839.02]	[2,187.10]	(164.19)
Percent White	64.35	64.46	-0.10	65.25	67.28	-2.03
	[18.98]	[25.79]	(1.85)	[19.06]	[20.66]	(1.56)
Percent High School Graduates	82.00	76.20	5.80***	82.90	82.50	0.40
	[16.39]	[21.11]	(1.51)	[16.32]	[19.92]	(1.49)
Median Household Income	92,189.30	55,779.55	36,409.74***	89,568.02	89,091.37	476.66
	[41,066.66]	[34,483.87]	(2,514.12)	[40,466.69]	[52,587.88]	(3,926.78)
Housing Units	1,818.58	1,727.90	90.67**	1,839.71	1,786.39	53.32
	[511.11]	[638.00]	(45.76)	[515.83]	[587.68]	(44.25)
Panel C: Locational Marginal Price (LMP, Main Outcome)						
Total LMP	39.68	37.30	2.38**	39.78	38.07	1.71
	[19.54]	[15.50]	(1.13)	[19.75]	[16.17]	(1.26)
System Energy Price	34.43	34.45	-0.03	34.44	34.45	-0.02
	[13.73]	[13.59]	(0.98)	[13.74]	[13.58]	(1.03)
Transmission-Induced Price	5.25	2.84	2.41***	5.34	3.62	1.73***
	[7.70]	[3.56]	(0.28)	[7.88]	[4.20]	(0.35)
Congestion Price	4.66	2.64	2.03***	4.75	3.23	1.52***
	[7.15]	[3.25]	(0.25)	[7.32]	[3.83]	(0.32)
Marginal Loss Price	0.59	0.21	0.38***	0.60	0.39	0.20***
	[0.67]	[0.52]	(0.04)	[0.68]	[0.55]	(0.04)
Number of Data Centers	1.23			1.23		
	[1.76]			[1.75]		
Number of Nodes	4.71	2.97	1.74***	4.74	3.08	1.66***
	[3.08]	[3.01]	(0.22)	[3.15]	[2.68]	(0.21)
Number of Unique Census Tracts	20	449		19	115	
Number of Census Tracts	20	449		19	190	

Notes: This table reports demographic characteristics and locational marginal prices (LMP) for treated and untreated census tracts before and after propensity score matching using ten nearest neighbors (10NN). The first year of data center entry in the estimation sample is 2017. To avoid using post-treatment information, matching is conducted based on demographic characteristics measured prior to 2017. Panels A and B report demographic characteristics in 2015 and 2016, respectively, before the first data center entry in the sample. These variables are used to construct the propensity scores. Panel C reports summary statistics for LMP components. Our primary outcome variable is the transmission-induced price component, defined as the sum of congestion and marginal loss prices. We also report summary statistics for congestion price, marginal loss price, total LMP, and system energy price. Total LMP is the sum of the system energy price and the transmission-induced price component. The bottom of the table reports the number of data centers, the number of pricing nodes within census tracts, and the number of census tracts before and after matching. Standard deviations are reported in brackets, and standard errors are reported in parentheses. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 2: Average Effect of Data Center Entry on Transmission-induced Price Component

	(1) Transmission-induced	(2) Congestion	(3) Loss
DC	2.4914**	2.3173*	0.1741***
	(1.2460)	(1.2047)	(0.0531)
Census Tract FE	×	×	×
Year FE	×	×	×
Observations	2,090	2,090	2,090

Notes: This table reports aggregated post-treatment estimates of the impact of data center entry on the transmission-induced price component, congestion price, and marginal loss price using the estimator of Sun and Abraham (2021). DC is an indicator equal to one if census tract i in year t has introduced its first data center. All regressions include census tract and year fixed effects. Because matching is conducted using ten nearest neighbors (10NN) with replacement, regressions are weighted by the inverse frequency of duplicated untreated tracts. Standard errors are clustered at the census tract level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Heterogeneous Effects of Data Center Entry on Transmission-induced Price Component: Generation Proximity and Fuel Type

	(1) Nearest-Generator Distance	(2) On-Tract Capacity	(3) Nearest-Generator Capacity
DC	0.2309	2.8469***	2.5494**
	(0.6889)	(0.9728)	(1.1288)
Dist (km)	0.0330		
	(0.0234)		
DC × Dist (km)	0.2739***		
	(0.0658)		
Fossil Fuel Capacity		-0.0013	-0.0026*
		(0.0014)	(0.0013)
DC × Fossil Fuel Capacity		-0.0034***	0.0001
		(0.0010)	(0.0051)
Renewable Capacity		-0.0294***	-0.0205**
		(0.0101)	(0.0081)
DC × Renewable Capacity		0.0342***	-0.0335
		(0.0108)	(0.0336)
Census Tract FE	×	×	×
Year FE	×	×	×
Adjusted Within-R2	0.023	0.016	0.018
Observations	2,090	2,090	2,090

Notes: This table reports heterogeneous effects of data center entry on the transmission-induced price component with respect to generation proximity and fuel type. The outcome variable is the transmission-induced price component, defined as the sum of congestion and marginal loss components of LMP. The treatment indicator equals one if census tract i in year t introduces its first data center. Column 1 examines heterogeneity by distance to the nearest generation resource. Distance (km) is a continuous variable measuring the distance between a census tract and its nearest generation facility. Column 2 examines heterogeneity by fuel type for generation located within the same census tract as the data center. Fossil Fuel Capacity measures average generation capacity from coal, natural gas, and oil, which are dispatchable resources. Renewable Capacity measures average generation capacity from solar, wind, and biofuels, which are intermittent and non-dispatchable resources. Column 3 examines heterogeneity by fuel type of the nearest generation resource, regardless of whether it is located within the same census tract. Fuel capacity variables are defined as in Column 2. All regressions include census tract and year fixed effects. Because matching is conducted using ten nearest neighbors (10NN) with replacement, regressions are weighted by the inverse frequency of duplicated untreated tracts. Standard errors are clustered at the census tract level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Heterogeneous Effects of Data Center Entry on Transmission-induced Price Component: Treatment Intensity and Demographics

	(1)	(2)	(3)	(4)
DC		2.132*	2.658*	4.097***
		(1.169)	(1.472)	(1.306)
# of DCs	0.805**			
	(0.376)			
DC × High Population		0.531		
		(1.699)		
DC × High HHI			-0.322	
			(1.857)	
DC × High White PCT				-3.140*
				(1.628)
Census Tract FE	×	×	×	×
Year FE	×	×	×	×
Adjusted Within-R2	0.015	0.013	0.013	0.018
Observations	2,090	2,090	2,090	2,090

Notes: This table reports heterogeneous effects of data center entry with respect to treatment intensity and demographic characteristics. The outcome variable is the transmission-induced price component, defined as the sum of congestion and marginal loss components of LMP. The treatment indicator DC equals one if census tract i in year t introduces its first data center. # of DCs is a continuous variable measuring the cumulative number of data centers in census tract i in year t. High Population indicates census tracts with above-median population based on 2015 data. High HHI indicates census tracts with above-median household income in 2015. High White PCT indicates census tracts with above-median share of White residents based on 2015 data. All regressions include census tract and year fixed effects. Because matching is conducted using ten nearest neighbors (10NN) with replacement, regressions are weighted by the inverse frequency of duplicated untreated tracts. Standard errors are clustered at the census tract level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

"The Hidden Cost of the Cloud: Data Centers and Transmission-Induced Electricity Prices"

Online Appendix (Not for Publication)

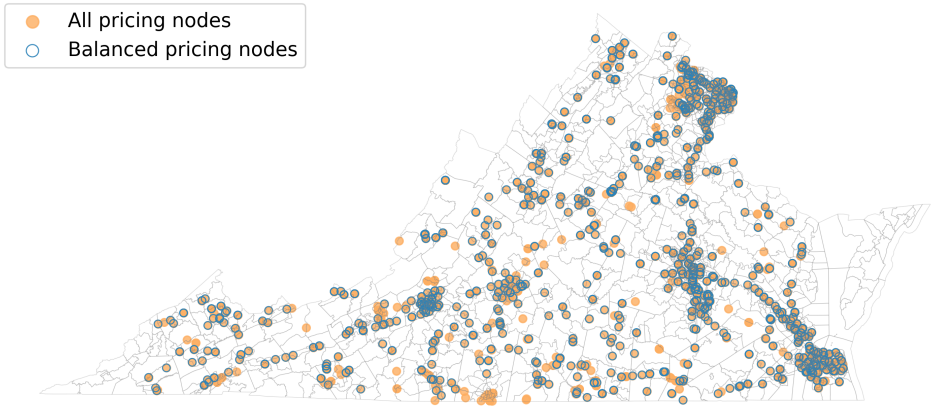
Jamal Mamkhezri
New Mexico State University
Georgia Institute of Technology

Xiaochen Sun
New Mexico State University

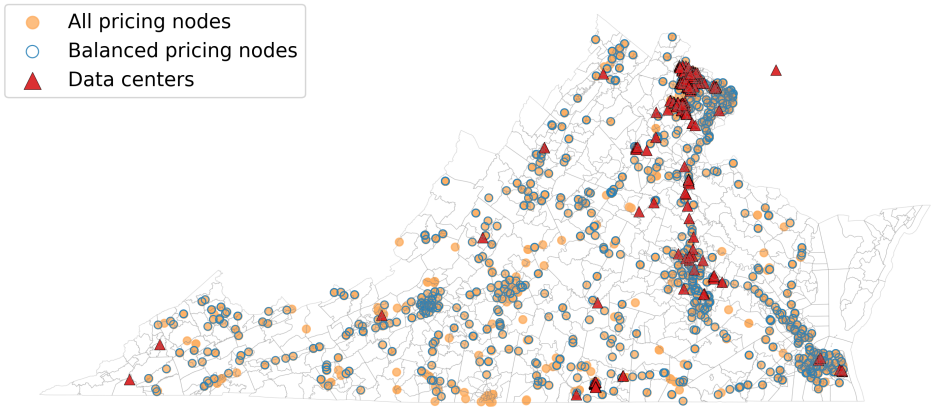
Yuting Yang
University of New Mexico

Appendix: Figures and Tables

Figure A.1: Pricing Nodes and Data Center Locations



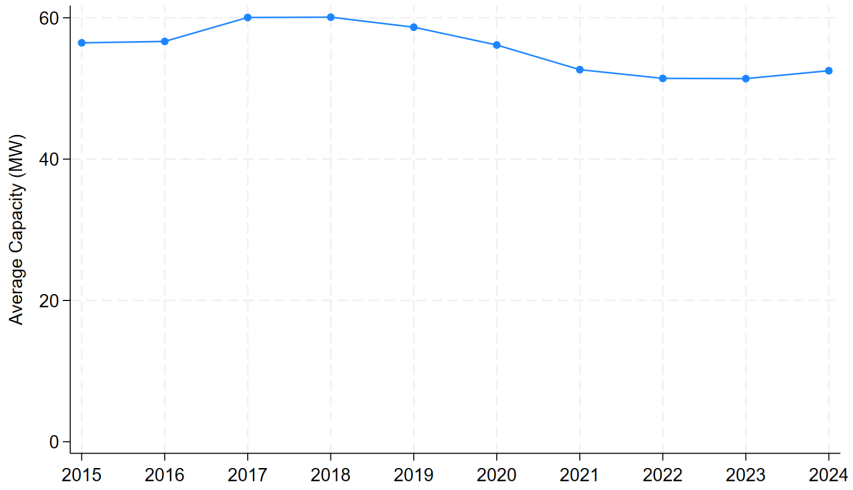
(a) Balanced vs. All Pricing Nodes



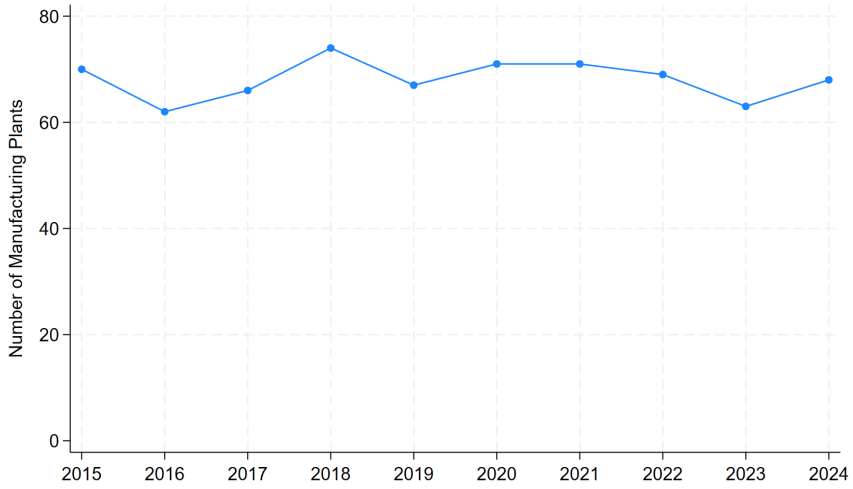
(b) Pricing Nodes and Data Center Locations

Notes: This figure shows the geographic distribution of pricing nodes and data center locations in Virginia. Panel a distinguishes between all pricing nodes observed during the study period and the subset of nodes that exist throughout the entire sample period (balanced nodes). Panel b overlays data center locations with pricing nodes in both categories.

Figure A.2: Supply- and Demand-Side Conditions in Virginia



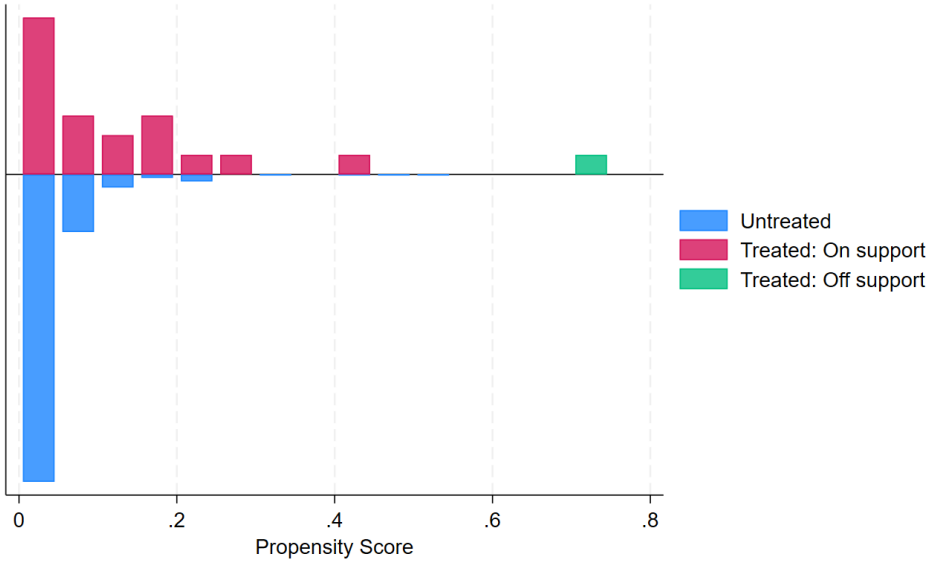
(a) Average Generation Capacity



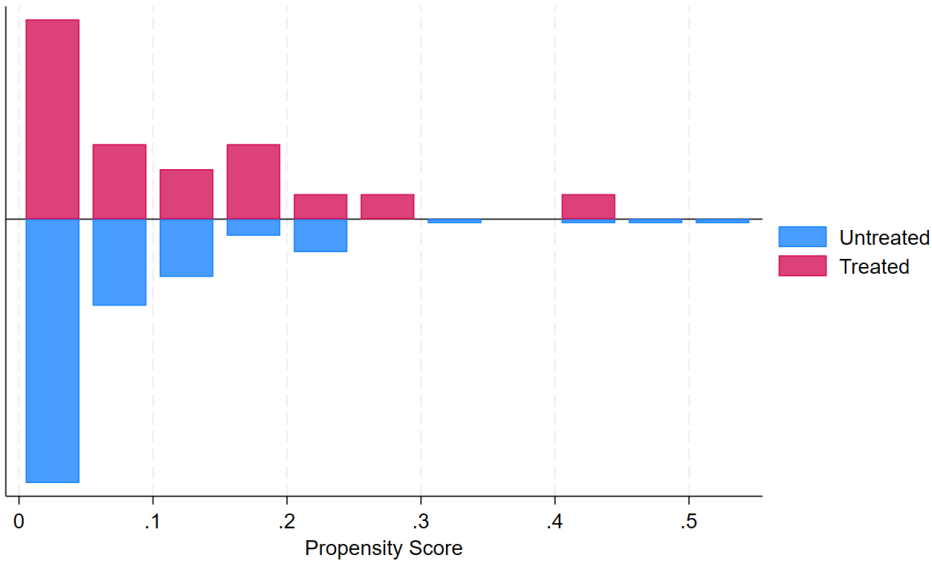
(b) Number of Manufacturing Plants

Notes: This figure presents descriptive evidence on supply- and demand-side conditions in Virginia during the study period. Panel a reports average generation capacity in Virginia over the study period. Panel b reports the number of manufacturing plants in Virginia based on the U.S. EPA Toxic Release Inventory (TRI). The figures illustrate that generation capacity and the number of major manufacturing facilities remain relatively stable over time, providing supportive evidence that large supply expansions or alternative demand shocks are unlikely to drive the pricing effects documented in the main analysis.

Figure A.3: Propensity Score Matching Based on 2015-2016 Demographic Characteristics
(10-Nearest Neighbor Matching)



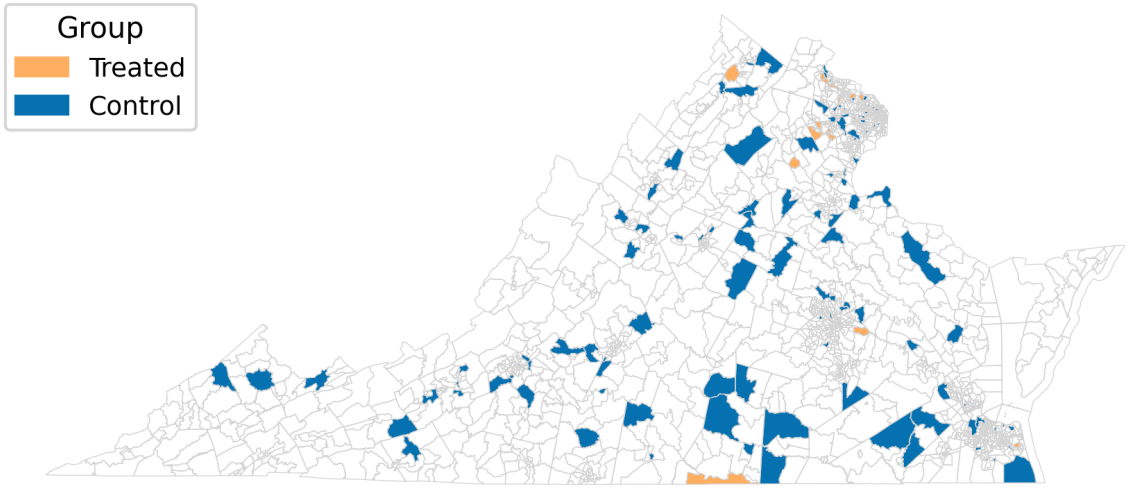
(a) Before Propensity Score Matching



(b) After Propensity Score Matching

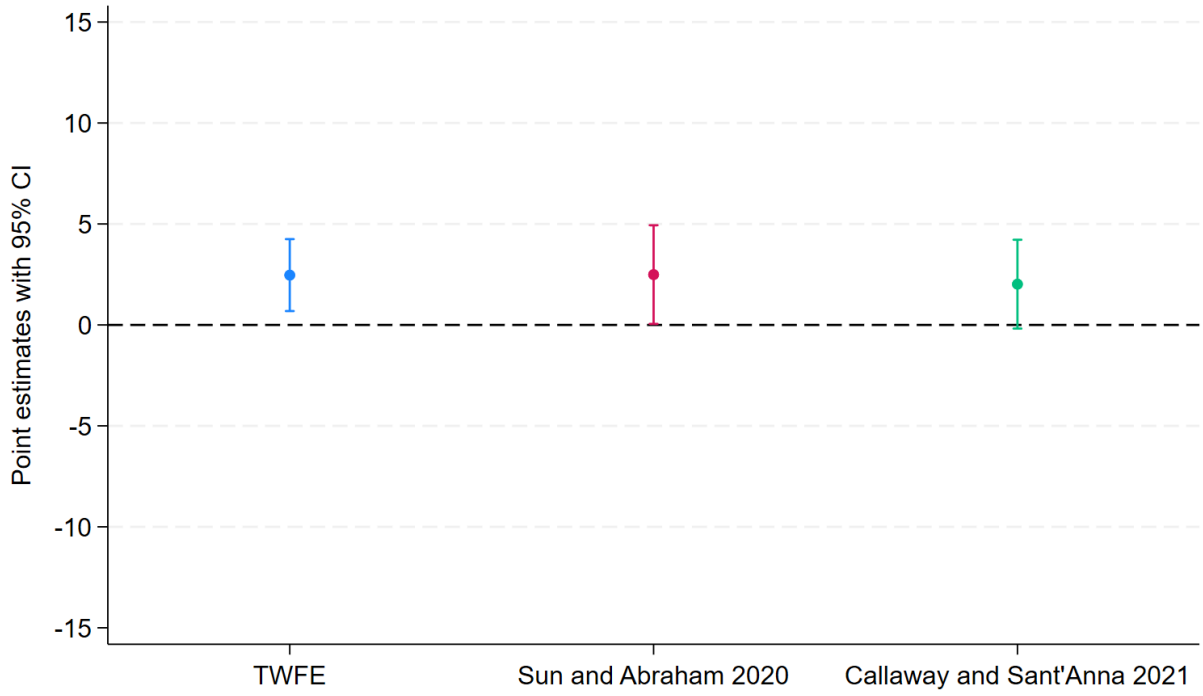
Notes: This figure presents the distribution of estimated propensity scores before and after matching using ten nearest neighbors (10NN). Red bars represent treated census tracts, and blue bars represent untreated census tracts. Green bars indicate treated tracts that are not matched to any untreated tract. Panel a shows the distribution prior to matching, and Panel b shows the distribution after matching.

Figure A.4: Locations of Treated and Matched Untreated Census Tracts



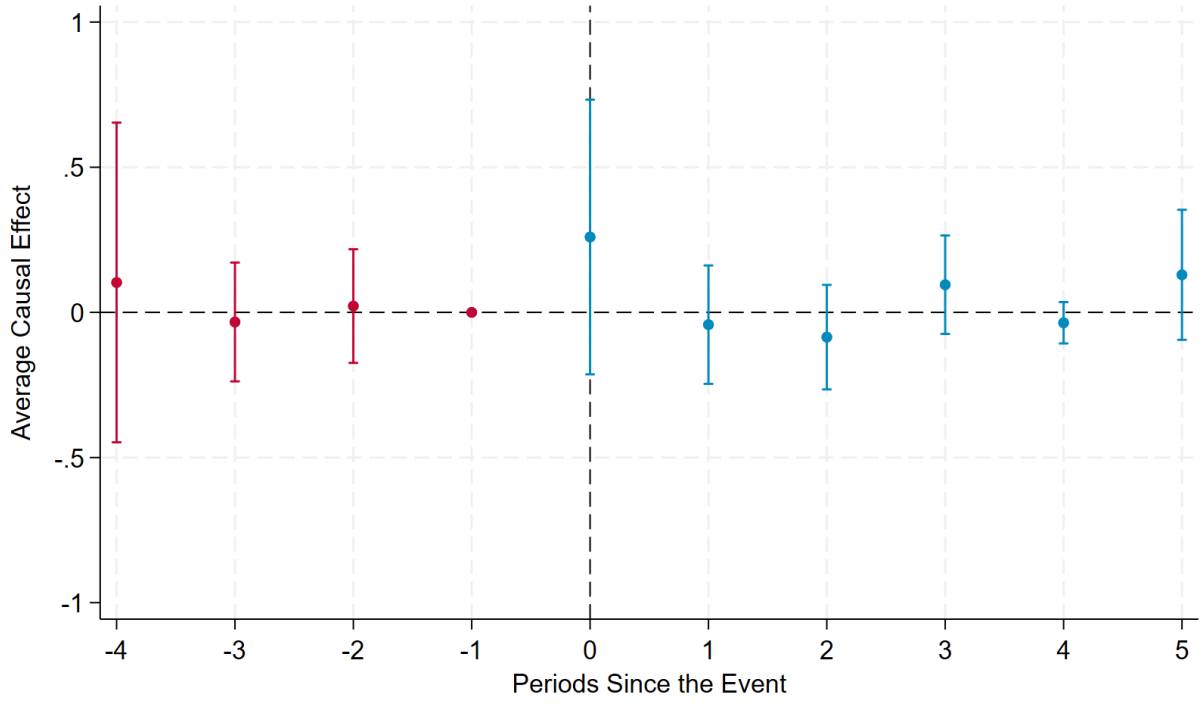
Notes: This map shows the locations of treated census tracts and matched untreated census tracts after propensity score matching using ten nearest neighbors (10NN).

Figure A.5: Comparison of Estimators: Transmission-induced Price Component

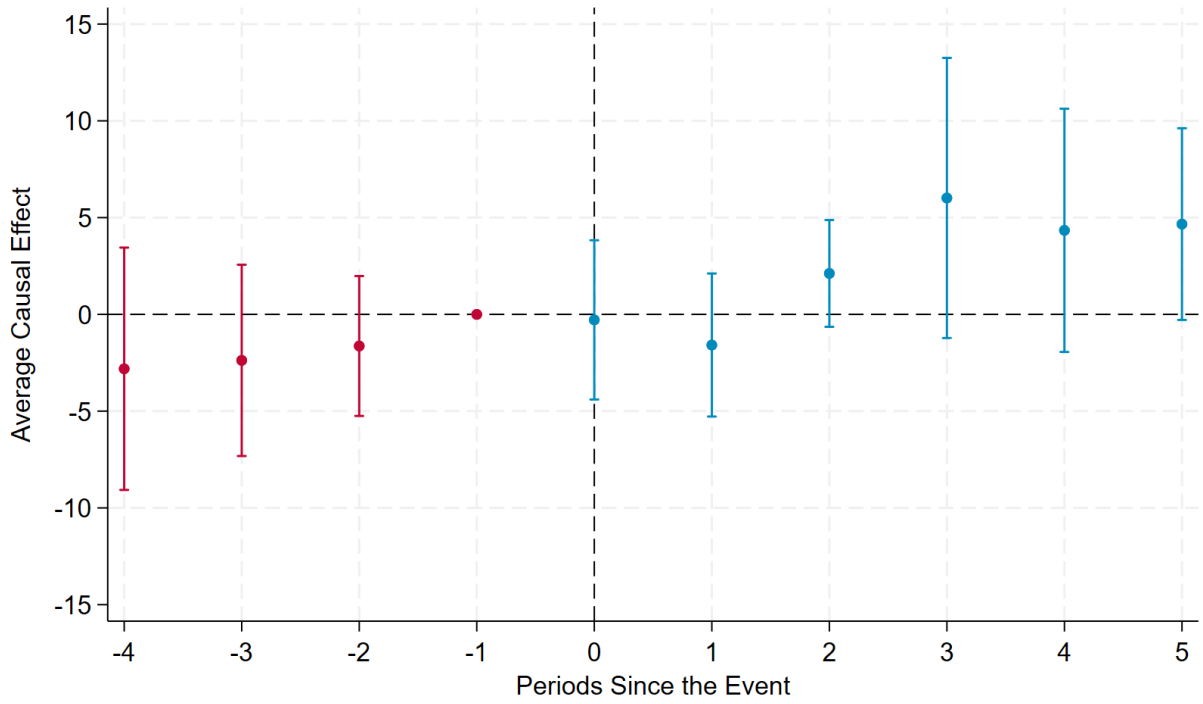


Notes: This figure compares aggregate post-treatment estimates of data center entry on the transmission-induced price component across alternative estimators: standard two-way fixed effects (TWFE), Sun and Abraham (2021) (our preferred estimator), and Callaway and Sant'Anna (2021). The outcome variable is the transmission-induced price component, defined as the sum of congestion and marginal loss components of LMP. Estimates are qualitatively similar across methods.

Figure A.6: Dynamic Effects of Data Center Entry on System Energy Price and Total LMP



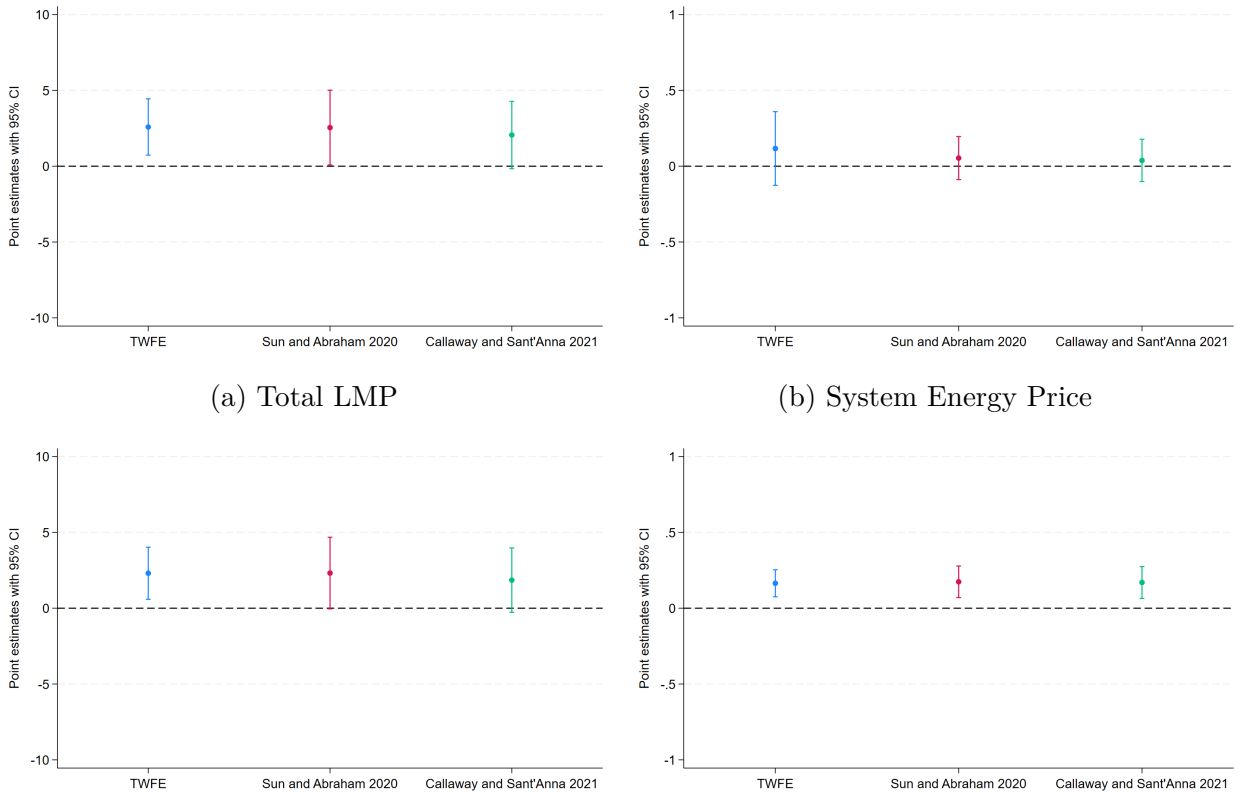
(a) System Energy Price



(b) Total LMP

Notes: This figure presents event-study estimates of the effect of data center entry on system energy price (Panel a) and total LMP (Panel b) using the estimator of Sun and Abraham (2021). Event times earlier than -4 and later than +5 are binned and not shown. All regressions include census tract and year fixed effects, and standard errors are clustered at the census tract level. System energy price shows no differential effect following data center entry, consistent with its uniform determination across pricing nodes within the same market. Total LMP increases beginning in year two and stabilizes after year three. Although most individual post-treatment coefficients are not statistically significant at the 5 percent level, the aggregated post-treatment effect is statistically significant (Appendix Table A.2).

Figure A.7: Comparison of Estimators Across LMP Components



(a) Total LMP

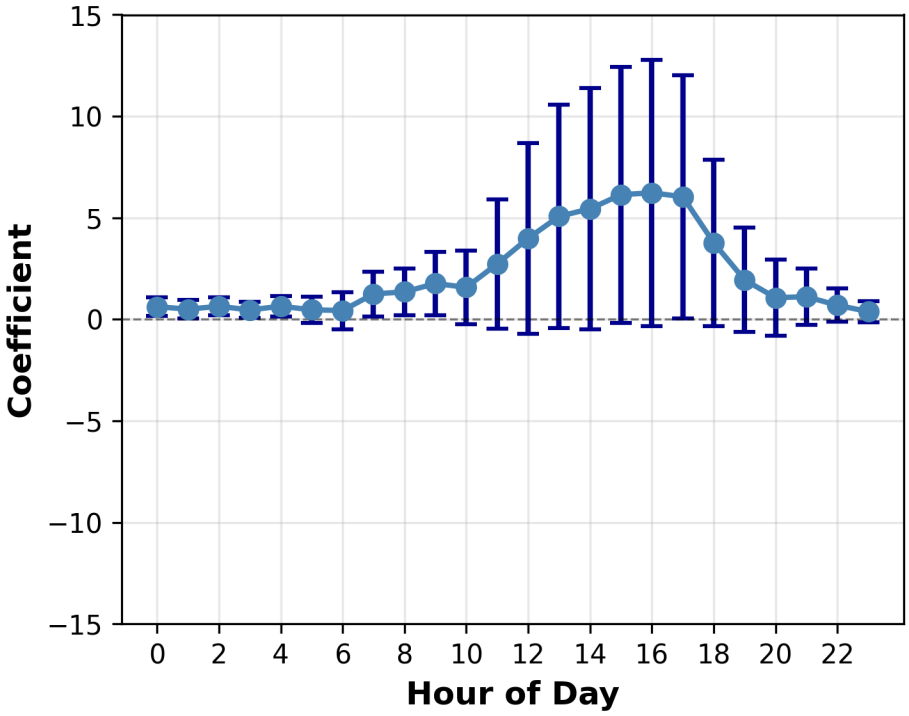
(b) System Energy Price

(c) Congestion Price

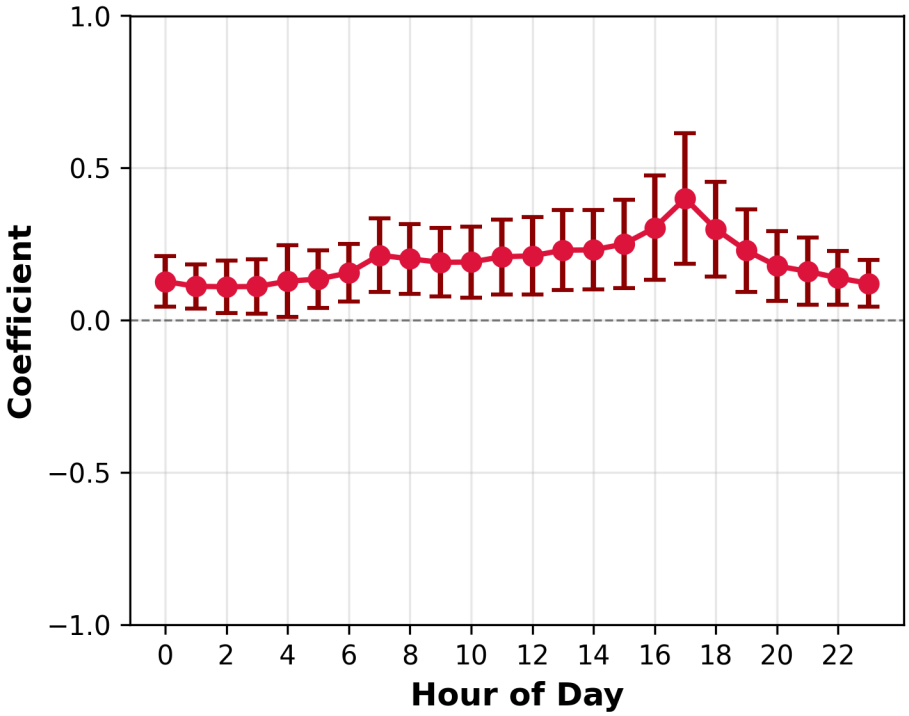
(d) Marginal Loss Price

Notes: This figure compares aggregate post-treatment estimates of data center entry across alternative estimators, standard two-way fixed effects (TWFE), Sun and Abraham (2021), and Callaway and Sant'Anna (2021), for total LMP and its components. Each panel reports the estimated average treatment effect for the corresponding outcome. Estimates are qualitatively similar across methods for all components.

Figure A.8: Impact of Data Center Entry on Congestion and Marginal Loss Prices by Hour



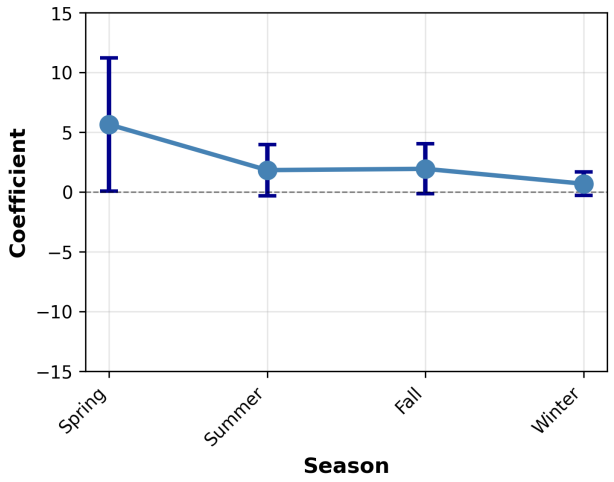
(a) Average Congestion Price



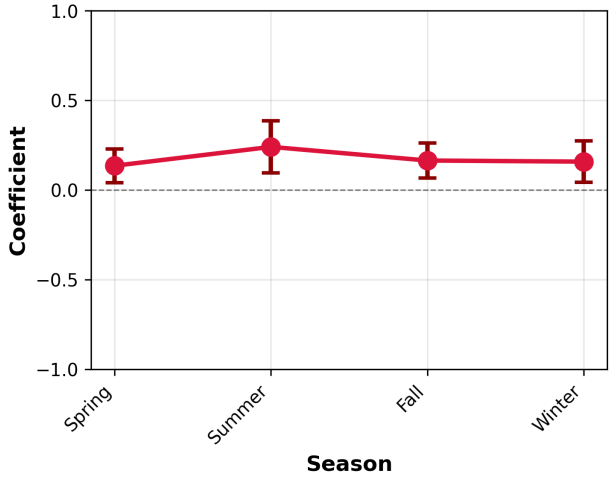
(b) Average Marginal Loss Price

Notes: This figure reports aggregate post-treatment effects of data center entry on congestion price (Panel a) and marginal loss price (Panel b) by hour of the day. Estimates are obtained using the estimator of Sun and Abraham (2021). Each point represents the average treatment effect over the post-treatment period estimated separately for each hour. Bars denote 95 percent confidence intervals. Standard errors are clustered at the census tract level.

Figure A.9: Impact of Data Center Entry on Congestion and Marginal Loss Prices by Season



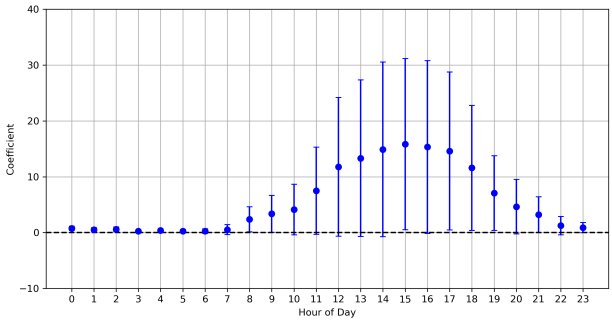
(a) Average Congestion Price



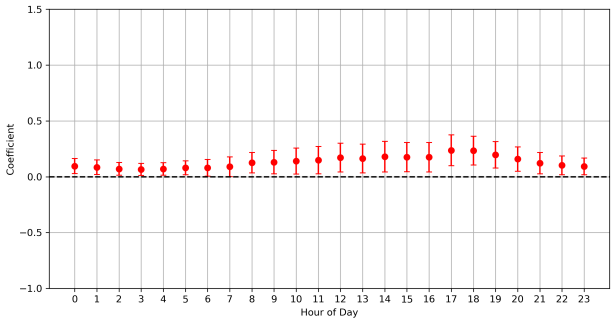
(b) Average Marginal Loss Price

Notes: This figure reports aggregate post-treatment effects of data center entry on congestion price (Panel a) and marginal loss price (Panel b) by season. Estimates are obtained using the estimator of Sun and Abraham (2021). Each point represents the average treatment effect over the post-treatment period estimated separately for each season. Bars denote 95 percent confidence intervals. Standard errors are clustered at the census tract level.

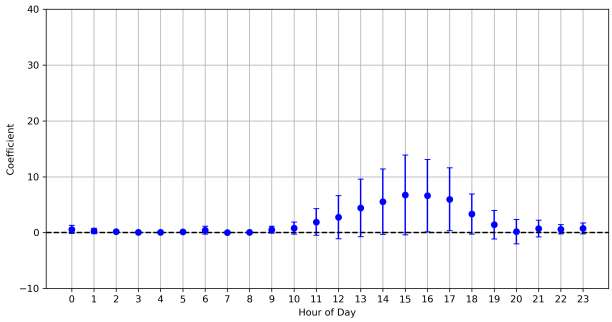
Figure A.10: Impact of Data Center Entry on Transmission-induced Price Component by Hour Across Seasons



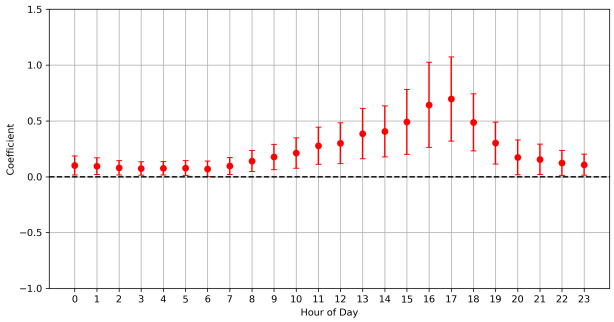
(a) Congestion Price, Spring



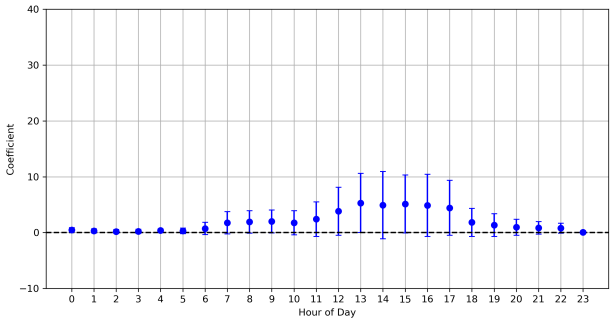
(b) Marginal Loss Price, Spring



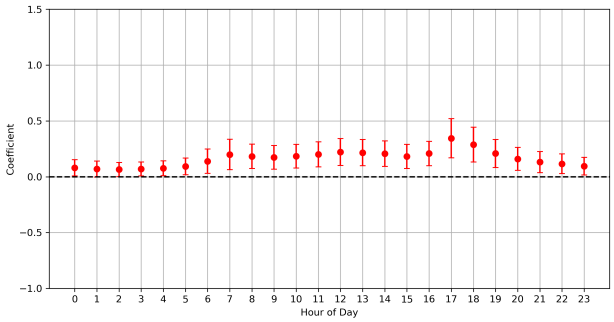
(c) Congestion Price, Summer



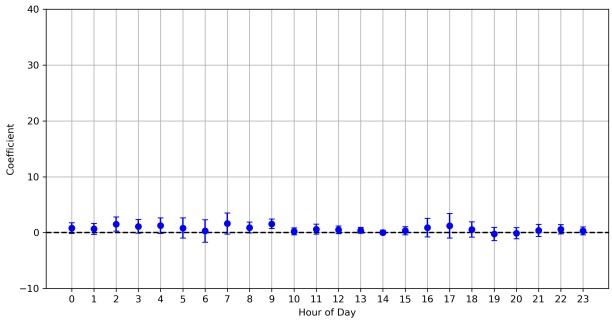
(d) Marginal Loss Price, Summer



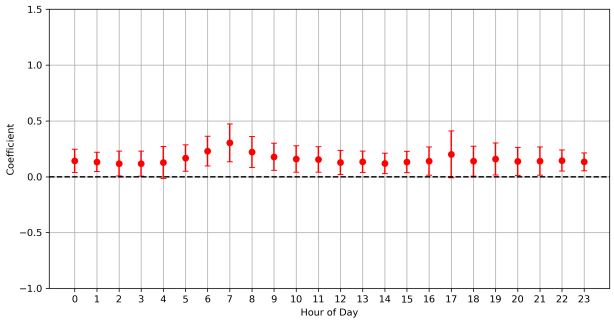
(e) Congestion Price, Fall



(f) Marginal Loss Price, Fall



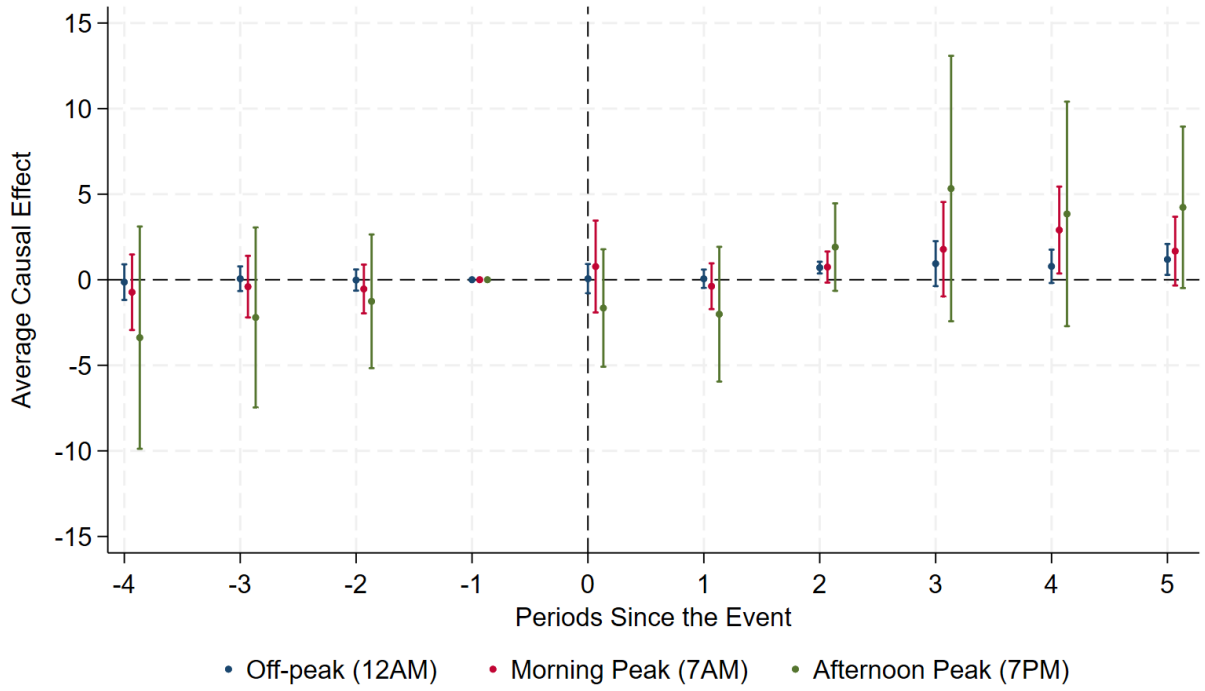
(g) Congestion Price, Winter



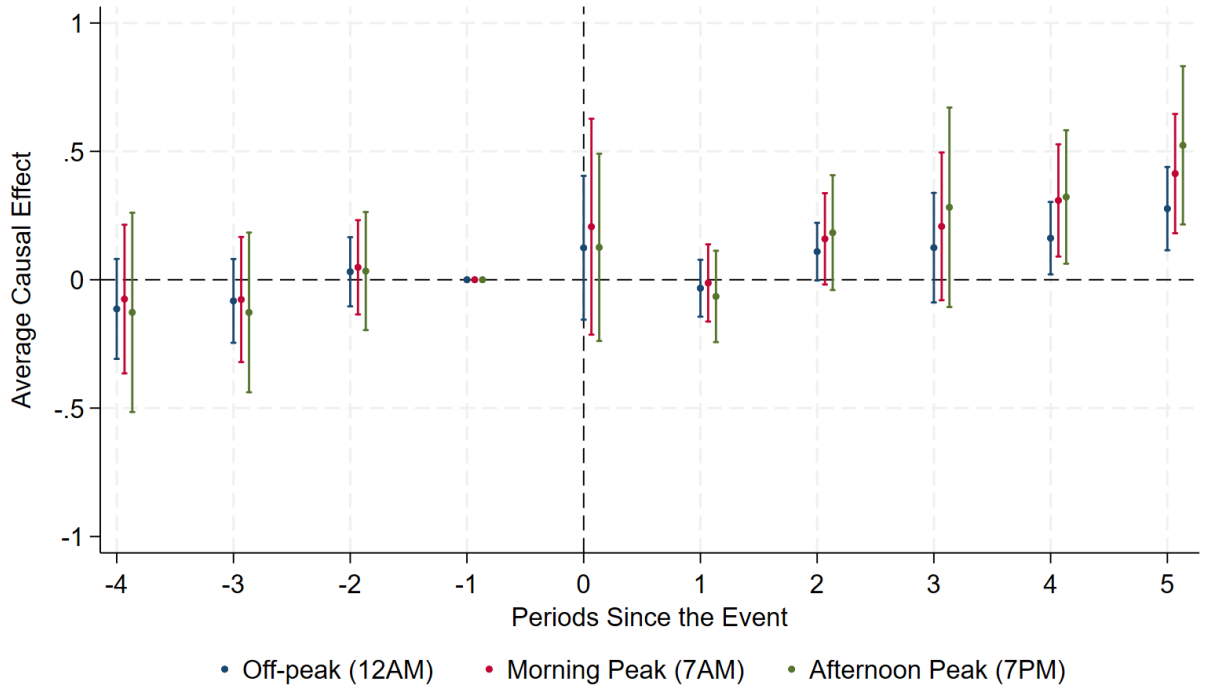
(h) Marginal Loss Price, Winter

Notes: This figure reports aggregate post-treatment effects of data center entry on congestion price and marginal loss price by hour within each season. Estimates are obtained using the estimator of Sun and Abraham (2021). Each point represents the average treatment effect over the post-treatment period estimated separately for each hour–season combination. Bars denote 95 percent confidence intervals. Standard errors are clustered at the census tract level.

Figure A.11: Dynamic Effects of Data Center Entry on Congestion and Marginal Loss Prices: Peak vs. Off-Peak



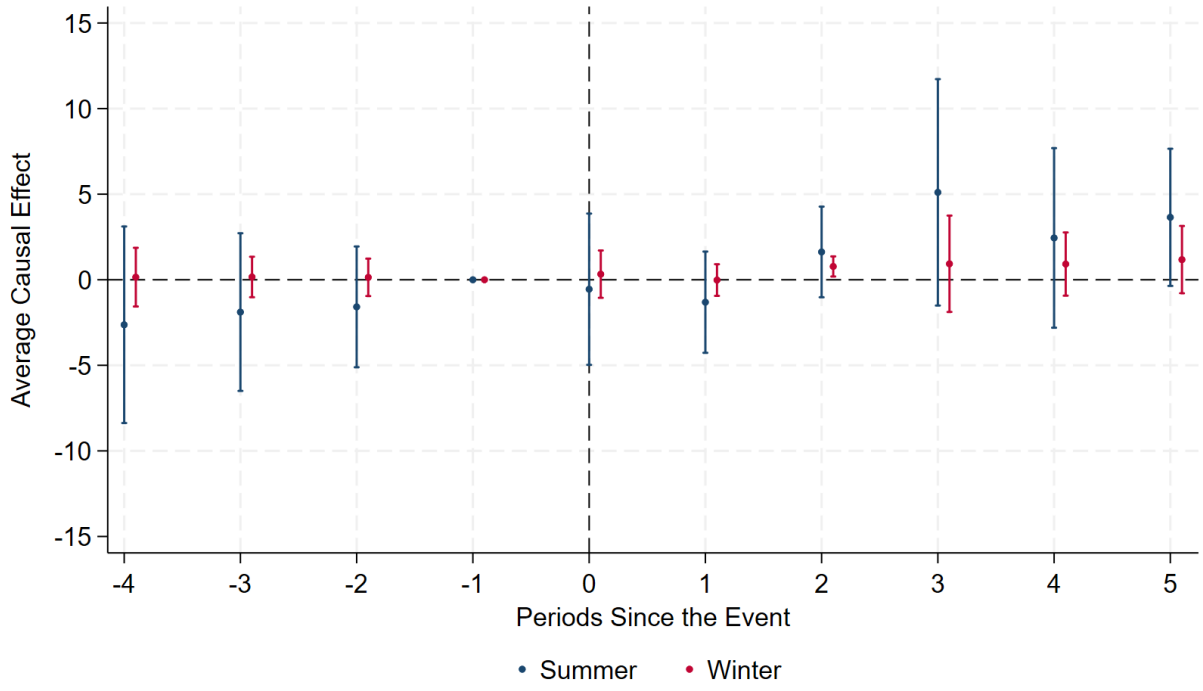
(a) Average Congestion Price



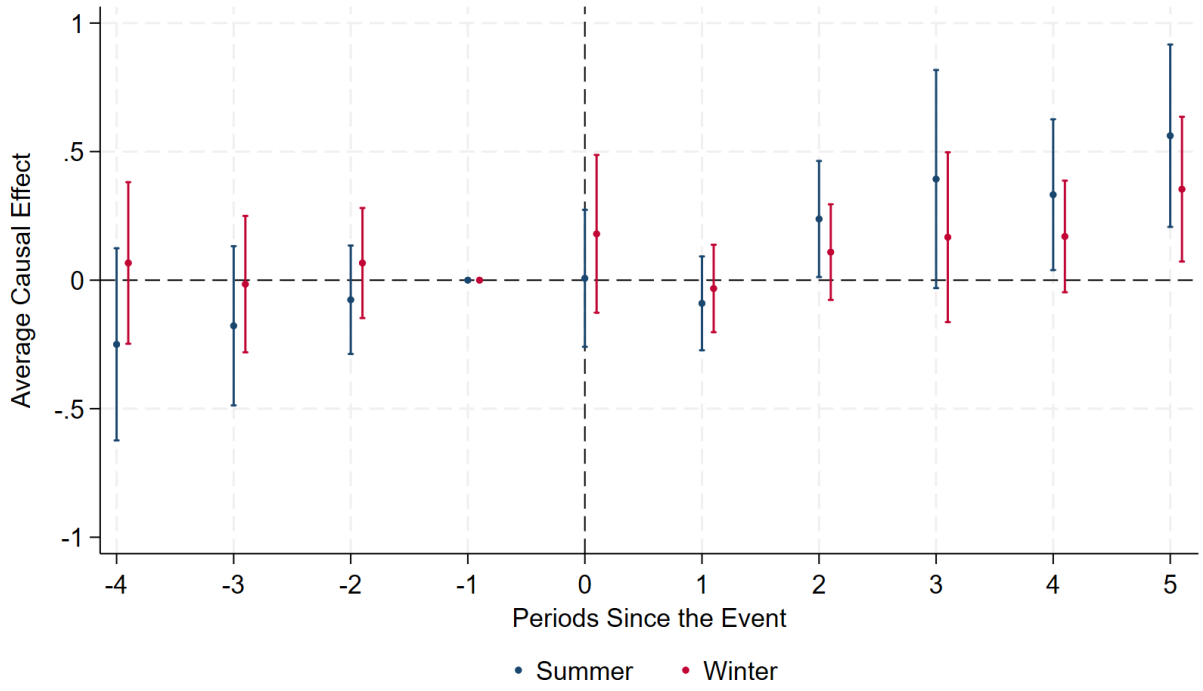
(b) Average Marginal Loss Price

Notes: This figure presents event-study estimates of the effect of data center entry on congestion price (Panel a) and marginal loss price (Panel b), separately for peak and off-peak periods. Estimates are obtained using the estimator of Sun and Abraham (2021). The reference year is defined as one year prior to the first data center entry in a census tract. All regressions include census tract and year fixed effects, and standard errors are clustered at the census tract level. Event times earlier than -4 and later than +5 are binned and not shown.

Figure A.12: Dynamic Effects of Data Center Entry on Congestion and Marginal Loss Prices: Summer vs. Winter



(a) Average Congestion Price



(b) Average Marginal Loss Price

Notes: This figure presents event-study estimates of the effect of data center entry on congestion price (Panel a) and marginal loss price (Panel b), separately for summer and winter. Estimates are obtained using the estimator of Sun and Abraham (2021). The reference year is defined as one year prior to the first data center entry in a census tract. All regressions include census tract and year fixed effects, and standard errors are clustered at the census tract level. Event times earlier than -4 and later than +5 are binned and not shown.

Table A.1: Correlation Between the Number of Pricing Nodes and Data Centers in a Census Tract

	(1)	(2)	(3)	(4)
# of DC	0.789***	0.675***	0.747***	0.585***
	(0.100)	(0.027)	(0.101)	(0.026)
Census Tract FE		×		×
Year FE			×	×
R2	0.013	0.969	0.017	0.973
Observations	4,690	4,690	4,690	4,690

Notes: The dependent variable is the number of pricing nodes in a census tract. # of DC is the number of data centers located within the tract. Standard errors are clustered at the census tract level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.2: Average Effect of Data Center Entry on Total LMP and System Energy Price

	(1) Total LMP	(2) Energy
DC	2.5449**	0.0535
	(1.2580)	(0.0723)
Census Tract FE	×	×
Year FE	×	×
Observations	2,090	2,090

Notes: This table reports aggregated post-treatment estimates of the effect of data center entry on total locational marginal price (Total LMP) and system energy price using the estimator of Sun and Abraham (2021). The treatment indicator equals one if census tract i in year t introduces its first data center. All regressions include census tract and year fixed effects. Because matching is conducted using ten nearest neighbors (10NN) with replacement, regressions are weighted by the inverse frequency of duplicated untreated tracts. Standard errors are clustered at the census tract level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.3: Generation Proximity and Fuel-Type Heterogeneity: Congestion and Marginal Loss Components

	Congestion Price			Marginal Loss Price		
	(1) Nearest Dist	(2) Tract Cap	(3) Nearest Cap	(4) Nearest Dist	(5) Tract Cap	(6) Nearest Cap
DC	0.0806	2.6778***	2.4055**	0.1503***	0.1691***	0.1440**
	(0.6617)	(0.9347)	(1.0900)	(0.0551)	(0.0490)	(0.0565)
Dist (km)	0.0249			0.0080***		
	(0.0225)			(0.0027)		
DC × Dist (km)	0.2710***			0.0028		
	(0.0614)			(0.0056)		
Fossil Fuel Capacity		-0.0012	-0.0027**		-0.0000	0.0001
		(0.0014)	(0.0013)		(0.0001)	(0.0001)
DC × Fossil Fuel Capacity		-0.0033***	-0.0001		-0.0001	0.0002
		(0.0010)	(0.0049)		(0.0001)	(0.0003)
Renewable Capacity		-0.0259***	-0.0201**		-0.0035***	-0.0004
		(0.0094)	(0.0079)		(0.0008)	(0.0007)
DC × Renewable Capacity		0.0290***	-0.0323		0.0052***	-0.0013
		(0.0102)	(0.0308)		(0.0011)	(0.0032)
Census Tract FE	×	×	×	×	×	×
Year FE	×	×	×	×	×	×
Adjusted Within-R2	0.023	0.015	0.018	0.017	0.013	0.012
Observations	2,090	2,090	2,090	2,090	2,090	2,090

Notes: This table reports heterogeneous effects of data center entry by generation proximity and fuel type, using congestion price and marginal loss price as separate outcome variables. Columns 1-3 use congestion price as the dependent variable, and Columns 4-6 use marginal loss price as the dependent variable. Columns 1 and 4 examine heterogeneity by distance to the nearest generation resource. Distance (km) is a continuous variable measuring the distance between a census tract and its nearest generation facility. Columns 2 and 5 examine heterogeneity by fuel type for generation located within the same census tract as the data center. Fossil Fuel Capacity measures average generation capacity from coal, natural gas, and oil, which are dispatchable resources. Renewable Capacity measures average generation capacity from solar, wind, and biofuels, which are intermittent and non-dispatchable resources. Columns 3 and 6 examine heterogeneity by fuel type of the nearest generation resource, regardless of whether it is located within the same census tract. Fuel capacity variables are defined as in Columns 2 and 5. All regressions include census tract and year fixed effects. Because matching is conducted using ten nearest neighbors (10NN) with replacement, regressions are weighted by the inverse frequency of duplicated untreated tracts. Standard errors are clustered at the census tract level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.4: Heterogeneous Effects of Data Center Entry on Congestion and Marginal Loss Prices: Treatment Intensity

	(1) Congestion Price	(2) Marginal Loss Price
# of DCs	0.738**	0.067***
	(0.363)	(0.018)
Census Tract FE	×	×
Year FE	×	×
Adjusted Within-R2	0.014	0.019
Observations	2,090	2,090

Notes: This table reports heterogeneous effects of data center entry by treatment intensity, measured by the cumulative number of data centers in a census tract. Column 1 uses congestion price as the dependent variable, and Column 2 uses marginal loss price as the dependent variable. # of DCs is a continuous variable measuring the cumulative number of data centers in census tract i in year t. All regressions include census tract and year fixed effects. Because matching is conducted using ten nearest neighbors (10NN) with replacement, regressions are weighted by the inverse frequency of duplicated untreated tracts. Standard errors are clustered at the census tract level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.5: Heterogeneous Effects of Data Center Entry on Congestion and Marginal Loss Prices: Demographic Characteristics

	Congestion Price			Marginal Loss Price		
	(1)	(2)	(3)	(4)	(5)	(6)
DC	1.977*	2.538*	3.840***	0.155***	0.120	0.258***
	(1.144)	(1.394)	(1.270)	(0.044)	(0.086)	(0.051)
DC × High Population	0.516			0.015		
	(1.648)			(0.077)		
DC × High HHI		-0.396			0.075	
		(1.781)			(0.095)	
DC × High White PCT			-2.959*			-0.180**
			(1.583)			(0.074)
Census Tract FE	×	×	×	×	×	×
Year FE	×	×	×	×	×	×
Adjusted Within-R2	0.012	0.012	0.017	0.010	0.011	0.014
Observations	2,090	2,090	2,090	2,090	2,090	2,090

Notes: This table reports heterogeneous effects of data center entry across demographic characteristics, using congestion price and marginal loss price as separate outcome variables. Columns 1-3 use congestion price as the dependent variable, and Columns 4-6 use marginal loss price as the dependent variable. The treatment indicator DC equals one if census tract i in year t introduces its first data center. High Population indicates census tracts with above-median population based on 2015 data. High HHI indicates census tracts with above-median household income based on 2015 data. High White PCT indicates census tracts with an above-median share of White residents based on 2015 data. All regressions include census tract and year fixed effects. Because matching is conducted using ten nearest neighbors (10NN) with replacement, regressions are weighted by the inverse frequency of duplicated untreated tracts. Standard errors are clustered at the census tract level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.6: Average Effect of Data Center Entry with Supply- and Demand-Side Controls

	Supply Side					Demand Side				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
	Total LMP	Energy	Transmission-Induced	Congestion	Loss	Total LMP	Energy	Transmission-Induced	Congestion	Loss
DC	2.5864***	0.0865	2.5000***	2.3345***	0.1655***	2.5841***	0.1168	2.4674***	2.3028***	0.1645***
	(0.9449)	(0.1065)	(0.9002)	(0.8679)	(0.0454)	(0.9420)	(0.1233)	(0.9037)	(0.8717)	(0.0454)
Total Capacity	-0.0001	0.0016	-0.0017	-0.0016	-0.0001					
	(0.0018)	(0.0016)	(0.0015)	(0.0015)	(0.0001)					
# Manufacturing Plants						0.0843	-0.0451	0.1295	0.2000	-0.0705**
						(0.3059)	(0.0296)	(0.2919)	(0.2769)	(0.0350)
Census Tract FE	×	×	×	×	×	×	×	×	×	×
Year FE	×	×	×	×	×	×	×	×	×	×
Adjusted Within-R2	0.013	0.064	0.013	0.013	0.010	0.013	0.002	0.013	0.012	0.014
Observations	2,090	2,090	2,090	2,090	2,090	2,090	2,090	2,090	2,090	2,090

Notes: This table reports average effects of data center entry while controlling for potential supply- and demand-side shocks. The outcome variables are total locational marginal price (Total LMP), system energy price, transmission-induced price component (defined as the sum of congestion and marginal loss components), congestion price, and marginal loss price. The treatment indicator equals one if census tract i in year t introduces its first data center. Total Capacity is a continuous variable measuring total generation capacity in census tract i in year t, addressing potential supply-side changes. # of Manufacturing Plants measures the number of manufacturing facilities in census tract i in year t, controlling for potential demand-side changes unrelated to data center entry. All regressions include census tract and year fixed effects. Because matching is conducted using ten nearest neighbors (10NN) with replacement, regressions are weighted by the inverse frequency of duplicated untreated tracts. Standard errors are clustered at the census tract level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.7: Average Effect of Data Center Entry Under Alternative Control Group Specifications

	(1) Total LMP	(2) Energy	(3) Transmission-Induced	(4) Congestion	(5) Loss
Panel A: Low Pre-treatment Correlated Tracts ($\rho < 0.5$)					
DC	3.3406**	0.0747	3.2661**	3.1354**	0.1307
	(1.4067)	(0.0908)	(1.4032)	(1.3324)	(0.1003)
Census Tract FE	×	×	×	×	×
Year FE	×	×	×	×	×
Observations	500	500	500	500	500
Panel B: W/O PSM					
DC	2.9451**	0.0383	2.9069**	2.6869**	0.2200***
	(1.2465)	(0.0663)	(1.2345)	(1.1907)	(0.0561)
Census Tract FE	×	×	×	×	×
Year FE	×	×	×	×	×
Observations	4,690	4,690	4,690	4,690	4,690
Panel C: W/O Weighting					
DC	2.2723*	0.0492	2.2231*	2.0630*	0.1601***
	(1.2434)	(0.0712)	(1.2309)	(1.1912)	(0.0510)
Census Tract FE	×	×	×	×	×
Year FE	×	×	×	×	×
Observations	2,090	2,090	2,090	2,090	2,090
Panel D: W/O Duplicated Untreated Tracts					
DC	2.5449**	0.0535	2.4914**	2.3173*	0.1741***
	(1.2727)	(0.0731)	(1.2607)	(1.2188)	(0.0539)
Census Tract FE	×	×	×	×	×
Year FE	×	×	×	×	×
Observations	1,340	1,340	1,340	1,340	1,340
Panel E: Matching W/ 5NN					
DC	2.2261*	0.0594	2.1668*	2.0008	0.1660***
	(1.2759)	(0.0760)	(1.2653)	(1.2234)	(0.0545)
Census Tract FE	×	×	×	×	×
Year FE	×	×	×	×	×
Observations	1,140	1,140	1,140	1,140	1,140

Notes: This table reports average effects of data center entry under alternative untreated group and matching specifications. The outcome variables are total locational marginal price (Total LMP), system energy price, transmission-induced price component (defined as the sum of congestion and marginal loss components), congestion price, and marginal loss price. The treatment indicator DC equals one if census tract i in year t introduces its first data center. Panel A reports results using an untreated group restricted to census tracts with low pre-treatment correlation in congestion prices relative to treated tracts, addressing concerns about potential spillovers. Panel B reports results using the full sample without propensity score matching. Panel C presents estimates without weighting by the inverse frequency of duplicated untreated tracts resulting from matching with replacement. Panel D reports results excluding duplicated untreated tracts altogether. Panel E shows results using an alternative matching procedure with five nearest neighbors (5NN). All regressions include census tract and year fixed effects. Standard errors are clustered at the census tract level. Significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.